

Social Projects and Neighbourhood Crime

Work in Progress

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October 5, 2026

Abstract

There is limited evidence on the effectiveness of community-based interventions at reducing local crime. Using quasi-random variation in the timing and location of micro-projects funded by the UK National Lottery, and a difference-in-differences approach, we find that communities receiving concurrent interventions totalling £30,000 or more experienced reductions in anti-social behaviour of up to 5.7 percent over a 24-month period. However, property and violent crime increase in parallel, so that there is no effect on overall crime. Larger infrastructure projects, similarly financed, significantly reduce violent crime by nearly 4%; an effect that is not offset by changes to property crime. Although these interventions provide new targets for criminals, suitable chosen and targeted interventions can play an important role in crime prevention strategies.

Keywords: Public Funding, Youth Crime, Place-based policy

JEL Codes: I38, J13, J14, K14, K42

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1 Introduction

Public policies aimed at preventing youth crime often focus on deterrent measures that increase the expected cost of crime for the offender. However, evidence on the effectiveness of increased policing and sanction threat remains ambiguous (Abrams (2012); Aizer and Doyle (2015); Hjalmarsson (2009); Kline (2012); Lee and McCrary (2017)). For this reason, many instead advocate for interventions that keep young people out of the streets (Beck, 2024). By reducing idle time, crime might fall mechanically in the short-run through an incapacitation effect (Anderson, 2014; Jacob and Lefgren, 2003). In the longer run, these interventions may, even more importantly, prevent the onset of a criminal career. Much evidence has shown the effectiveness of education, the location where young people (should) spend most of their time, as a key crime-prevention strategy (Hjalmarsson et al. (2015); Lochner and Moretti (2004); Machin and Meghir (2004)). Such education expansions combine increased incapacitation with higher human capital accumulation which increases the opportunity cost of a criminal career (Bell et al., 2022; Lochner, 2020). Yet, bringing more youths together, particularly those with a higher propensity for criminal behaviour, can also lead to higher exposure to criminal peers and geospatial concentration of criminal activities (Akee et al., 2014; Steinberg et al., 2019).

This paper provides evidence on the impact of community-led interventions, mostly targeting young people, on local crime. These bring youths together outside home and school contexts, and have received little attention in the literature to date. These interventions are funded by the UK National Lottery Community Fund and by Sport England (NLCF/SE). NLCF and SE distribute funds, raised by a levy on the National Lottery, to community-led organisations to help improve their communities, encourage participation in sporting activities, boost skills and employability of those out of work and provide mental health support to locals that need it. While the funds are not solely focused

on youth, or specifically targeting crime, we concentrate our investigation on projects which are the most likely to benefit young-people. The NLCF/SE funds about 10,000 projects per year, across two schemes: micro projects (less than $\leq$ £10,000) and larger projects (up to several million pounds), which we investigate separately.

We exploit quasi-random local variation in the timing of funding to provide causal inference on the impact of these projects on neighbourhood-level reported crimes. We estimate impacts at the extensive and intensive margin, differentiate between project size, type, and target audience, and examine treatment effect heterogeneity by neighbourhood characteristics.

We use a two-stage difference-in-difference approach that compares the evolution of crime rates in areas exposed to at least one micro project with those that were unaffected over the same period (Gardner, 2022). At the extensive margin, a micro project reduces anti-social behaviour by about 1% relative to the monthly average, but at the same time increases property crime by the same magnitude, so no effect on overall crime is detectable. Since neighbourhoods can be the recipients of funding of more than one project concurrently, we also estimate the intensive margin of funding. We find a similar shift in the composition of criminal activity, with no detectable effect on overall crime: Neighbourhoods receiving £30,000 or more in simultaneous micro projects experience a reduction in anti-social behaviour (ASB) of 5.7%, but also an increase in non-ASB crimes by 3.4% over a 24 months period. These results are context-dependent: urban areas and those with lower levels of education see reductions in ASB of around 7-8%.

Finally, we find that the aims and objectives of community-led projects matter. Crucially, an additional £10,000 worth of micro projects aimed at improving the ‘Skills & Opportunities’ of youths reduces ASB by nearly 2% without causing

an increase in property crime, akin to the large body of evidence showing that the acquisition of skills through education acts as a crime deterrent.

In the second part, we consider the impact of larger projects, which could be worth several millions over a multi-year horizon. We find a weakly significant reduction in overall crime at the extensive margin of just under 1%. This is primarily driven by a nearly 4% fall in violent crime, which is offset partly by growing property crime. In contrast to micro projects, where increasing the number of active projects leads to stronger effects, we do not find a robust relationship between funding and impact for large projects. In addition, we do not find evidence of spatial spillover effects, such as crime location displacements, for either small or large projects.

Consistent with the micro intervention estimates, we find that violent crime falls most in urban areas and those with high shares of young residents. Most importantly, it is again the ‘Skills & Opportunities’ that provide the highest return; they decrease violent crime by 7% - without offsetting increases in property crime.

By providing the first causal estimates of community-led projects, we advance the understanding of the efficacy of place-based crime prevention policies. The literature on the effectiveness of non-policing place-based intervention is generally positive. Branas et al. (2018) or Chalfin et al. (2021), using RCTs, report that greening vacant lots or street-lighting substantially reduces crime in Philadelphia and New York, respectively. General improvement in neighbourhoods, such as those funded by voluntary tax paid by local businesses reduces crime by about 10% in the US and UK (Cook and MacDonald, 2011; Faggio, 2022). However, Blattman et al. (2021) reported no effect on crime over an 8-months period of an RCT randomising the allocation of public services in Bogotá. Other interventions, involving young people directly have also generated

crime reduction. For example, Heller et al. (2017) reports estimate from RCTs providing behavioural training to reduce violence and report a drop in crime of about 30%. We focus on small-scale projects that do not specifically emphasise crime reduction, but may indirectly, either due to incapacitation effect or change in local norms (Acemoglu and Jackson, 2017). Many of these projects involve sport either as a participative event or infrastructure. While Nichols (2010) suggests that sport programmes form an integral part of a national crime rate reduction strategy, empirical evidence on their effectiveness, particularly among youth, has been mixed (Jugl et al., 2023).

The closest work to ours is Villa (2024), who estimate the effect of youth-specific community programs in London. She uses a difference-in-difference strategy and plausibly exogenous reduction in public funding driven by austerity measures to estimate that young people became 15% more likely to be engaged in criminal activities after the closure of youth centres. Like our programs, youth centres do not specifically target crime reduction, but might affect engagement in crime with a mix of incapacitation and activities that fosters human capital and social interactions. Our findings complement this research by investigating a broader range of low intensity interventions throughout the country. The national spread of these community-led programmes and the diversity of funded interventions allows us to investigate heterogeneity in the effect by wards characteristics and by type of projects.

The rest of the paper is structured as follows. In Section 2, we provide background information on the National Lottery Funds and the funding allocation. Section 3 describe the various datasets. Section 4 introduces the identification strategies: two-stage difference-in-difference for micro- and large projects, and describes tests for the validity of approaches. Sections 5.1 and 5.2 discuss the results for small and large projects, respectively, including several heterogeneity and robustness checks. Section 6 concludes and provides a discussion on the

policy relevance of the findings.

2 The National Lottery Community Fund

The National Lottery is one of the largest funders of community projects and charitable causes in the UK. Since its launch in 1994, the franchise has invested £45.5 billion in funding to over 672,000 projects country-wide.¹ According to the franchise contract with the UK government, 25% of the National lottery revenues have to fund ‘good causes’. This levy is administered by the public National Lottery Distribution Fund, which distributes them through 12 independent distributors that differ in their objectives respectively geographic coverage; We focus on two of these distributions: the *National Lottery Community Fund*, which receives about 40% of the total funds, and Sport England, the main distributor of money to fund sports (herein, NLCF/SE) that has received just under 25%. They act as funding bodies to charitable organisations, clubs and societies aiming to improve their local communities across England and Wales.

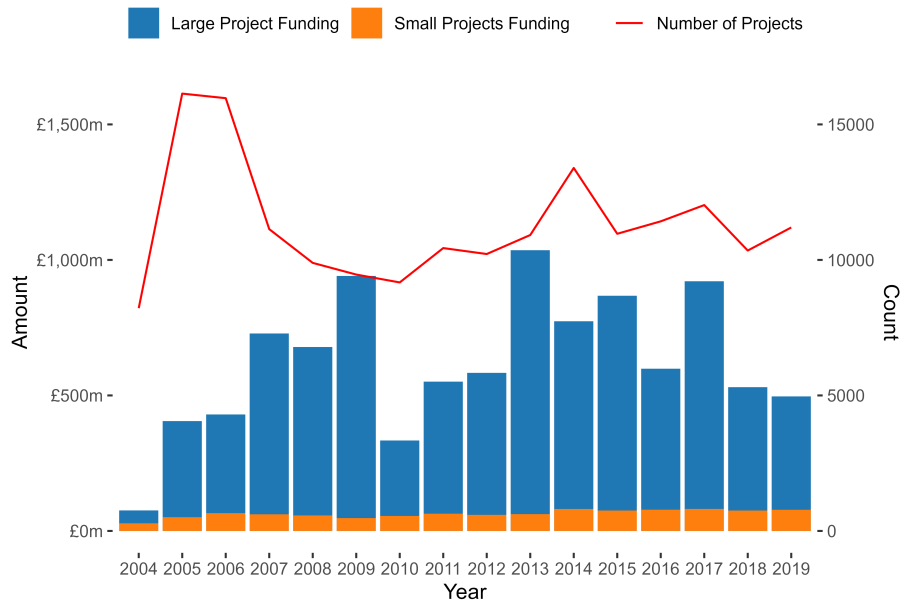
Charitable organisations apply directly to the NLCF/SE for a grant to support a local project.² Applications need to be made at least 16 weeks before the anticipated start date of the project, and are evaluated by a regional funding officer, who usually oversees one or more local authority districts in a given region. Overall, approximately 48% of applications are successful. NLCF/SE supports various funding streams that can be split into two main categories: micro projects worth up to £10,000, and larger projects worth up to £5m. Both the application process and regulations governing the types of applications that can be made differ between the two. Micro projects have a fixed duration of 24 months, and undergo less rigorous review. Larger projects, in contrast, are typically obtained by larger charities and voluntary private organisations. They are

¹For more information see the latest annual report for 2024-25.

²There are strict rules that regulate eligibility. For example, no individuals can be funded as the opportunity is restricted to established groups and organisations. For more information, see the ‘Who can apply?’ page.

expected to last around 5 years, though in some cases can be active for as long as 10 years. The total funding and number of projects awarded over the period 2004 to 2019 are illustrated in Figure 1. The number of funded projects varies substantially from year to year from a low of 9,102 in 2009 to a high of 16,100 in 2005. The total spend varies from £400 million to £1000 million, mostly driven by variations in awards to large projects. Each year, micro projects represent 80 to 90% of the total number, yet large projects account for about 90% of the overall budget.

Figure 1: Annual Total Funding and Project Counts (2004-2019)



Notes: Time series plot for the full sample of projects (youth and non-youth). Projects are assigned to each calendar year based on the start-date of project (not award date). For multi-year projects, the funds and number of projects is only reported for the first year. The plot begins in 2004 due to only a small number of projects being recorded prior.

All projects have a common aim to “bring people together”, or to “improve the places and spaces that matter to communities”. Typically, projects are oriented towards deepening ties among the local community, supporting group activities, and improving well-being in the local area. This can be achieved by, for example, one-off events or the funding of equipment or buildings. Projects

vary by size, type and targeted population. The following examples may help illustrate these differences:

- In June 2007, an award of £7408 was made to the Northampton School for Girls to fund an extra curricular indoor athletics club to provide better opportunities for their students and local primary schools. The grant covered fees, materials and equipment.
- In April 2019, the community in Wilmslow were granted £10,000 for 12 months to "organise a Redeeming Our Community (ROC) Cafe for young people between the ages of 11-18. The aim of the project was to provide opportunities through various youth club activities for vulnerable teenagers, to make friends, socialise, and find necessary mental health support." The funding covered the operating costs of the cafe for 12 months.
- In January 2017, Blackpool Council were awarded £10,379,790 for a 5-year project under the "HeadStart" funding programme, to "explore and test new ways to improve the mental health and wellbeing of young people aged 10 to 16 and prevent serious mental health issues from developing".

3 Data

3.1 Funded Projects

Data on funded projects was extracted from several sources, largely drawing from the two most important distributors of National Lottery funds: the National Lottery Community Fund and Sport England (for a full list, see Table A.1 in the Data Appendix). We extract projects granted during the period 2001 to 2019 to exclude the COVID-19 pandemic period, when mobility restrictions affected both the operations of charities and crime. This results in 292,717 unique records. Each record contains the name of the recipient organisation, the funded amount, the start date, location at the ward-level and description of the project. To reduce noise and measurement error, we exclude projects

that are worth less than £1,000, and focus on projects in England and Wales only since the crime data are not available for Scotland and Northern Ireland.³ This yields a sample of 180,915 records. Next, the project description is used to classify projects according to the demographic group they target. Specifically, we classify projects into youth and non-youth-oriented ones using a set of key words applied to the descriptions of the projects.⁴ “Youth” projects either target young people directly, or benefit the wider community including young people. “Non-youth” projects subsume all other projects. Given the strong age gradient in criminal offending, we do not expect the latter group to affect crime outcomes.⁵ Approximately 48% of all projects fall into the “Youth” category and constitute the main sample for analysis, leaving a final sample of 87,599 projects.⁶

We geolocate project locations to (electoral) wards⁷, the primary unit of English electoral geography. A ward typically represents about 5,500 electors, but ward population counts and acreage can vary substantially.⁸ For the rest of the manuscript, we will refer to ward and neighbourhood interchangeably. There is considerable variation in the quantity of awarded projects across neighbourhoods, as illustrated in Figure 2. It splits wards into percentiles according to the number of projects they received throughout our analysis period. The median ward receives 7 projects, while those at the 10th and 90th percentile had projects funded 4 respectively 22 times. Geographically, urban areas are over-represented at higher percentiles, though this is likely driven by differing numbers of eligible organisations operating in these neighbourhoods.

³For a detailed step-by-step overview of the data extraction and cleaning procedure, including a discussion of how miscoded location information was handled, see the Data Appendix.

⁴For more detail on how this is achieved see the discussion in Appendix A.1.

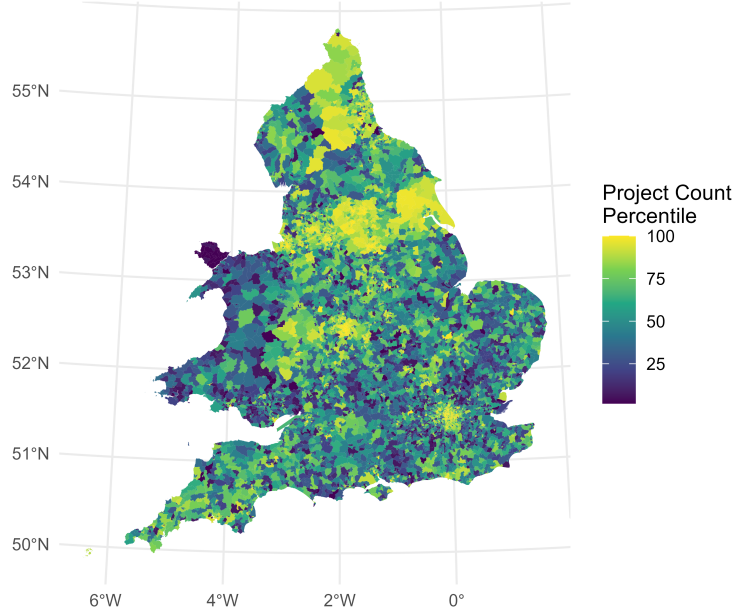
⁵See the set of robustness checks in section (4) where this assumption is verified.

⁶For more information on classification procedure and a set of summary statistics see Appendix A.1.3.

⁷For details on this procedure, see Appendix A.1.2.

⁸In 2011, there were 8,546 wards in England and Wales. Periodically, boundary adjustments are made to keep ward’s sizes relatively constant. For consistency, we define all wards based on their 2011 borders using the 2011 Census Merged Wards (CMWD).

Figure 2: Heat Map of Percentiles of the Number of Projects (2011-2019)



Notes: Values are the percentile of the total number of projects received by each ward over the 2011-2019 period.

In addition, we use the project description to classify projects according to their content and objectives. First, we consider four broad categories: “Health & Well-being”, “Skills & Opportunities”, “Civil & Social Engagement”, and “Other”. This type differentiation is useful to investigate the mechanisms at play, and helps to pinpoint which type of intervention may be most effective at reducing crime. For example, “Health & Well-being” projects may help young people regulate aggression, thus reducing ASB crimes, and may be related to substance crime, while “Skills & Opportunities” projects should improve the wage prospects in the legal economy. Second, we further classify by whether they are long or short-term oriented: “Assets & Infrastructure”, “Activities & Services”, and “Other”. Summary statistics for projects across both size and type are given in Table 1 below.

Table 1: Descriptive Statistics by Project Group

Group	N	Median (£)	SD (£)
Overall	87599	9680	536411
Micro	68649	8259	2820
Large	18950	95980	1126175
Health & Well-being	50494	9950	611925
Skills & Opportunities	15320	8546	211589
Civil & Social Engagement	2010	9492	110859
Other Activities	19775	8452	529163
Assets & Infrastructure	15577	10000	626175
Activities & Services	57029	9580	430087
Other	14993	9210	754952

^a Note: Values are given to the nearest pound sterling.

Projects over £10,000 are classified as Large; those worth £10,000 or less are Micro.

Lastly, we emphasise the difference between small and large projects that plays an important role in our empirical analysis. The median micro project has an award value of about £8,259, while large project budgets amount on average to £95,980.⁹

3.2 Local Crime

The source of recorded crime is the police crime records compiled for the 43 Police Force Areas (PFAs) in England and Wales.¹⁰ The raw data contains the monthly counts for 15 different types of crime at the Lower Super Output Areas (LSOAs) level, equivalent to a census track.¹¹ Recorded crimes are available from January 2011, and we stop the collection in December 2019 to exclude the COVID-19 pandemic period. We aggregate these data to the electoral ward-level using 2011 ward code identifiers to create a time-consistent geography. Appendix A.2 describes the aggregation process and linking procedures in de-

⁹For a split of these projects by size and type see Table B.2 in Appendix B

¹⁰Available at <https://data.police.uk>. Data from the British Transport Police has been excluded due to potential measurement error associated with the location of the reported crimes.

¹¹Though crime is recorded at the street-level, to preserve anonymity, the publicly available data is recorded at the LSOA-level, a slightly coarser geography. Crime reports are available for 35,750 out of the 35,753 LSOAs that are active during this period. The remaining three LSOAs have almost no reported crime. Moreover, data is not available for all LSOAs that fall into the Manchester PFA after July 2019 due to misreporting.

tail.

To reduce measurement error from classification changes over time, we aggregate crimes into 6 broad groups:¹² i) ‘total crime’, ii) ‘total crime excl. ASB’ (the sum of all crime excluding anti-social behaviour), iii) ‘ASB’ (anti-social behaviour), iv) ‘property crime’ (burglary, shoplifting, vehicle crime, bike theft, persons theft and other theft); v) ‘violent crime’ (violent & sexual offences, robbery, arson and criminal damage, possession of weapons, public order and disorder incidents), and vi) ‘substance crime’. Changes in Home Office Counting Rules (HOCR), that took effect in April 2018, for violence against persons and crimes relating to stalking and harassment, imply that after that date violent crimes are no longer comparable to the rest of data. Hence, we discard the affected periods for ‘total crime’, ‘total crime without ASB’ and ‘violent crime’. Table 2 provides summary statistics for these six crime categories, expressed per 1,000 residents per month at ward-level, based on the 2011 Census population.

Table 2: Broad Crime Categories Summary Statistics

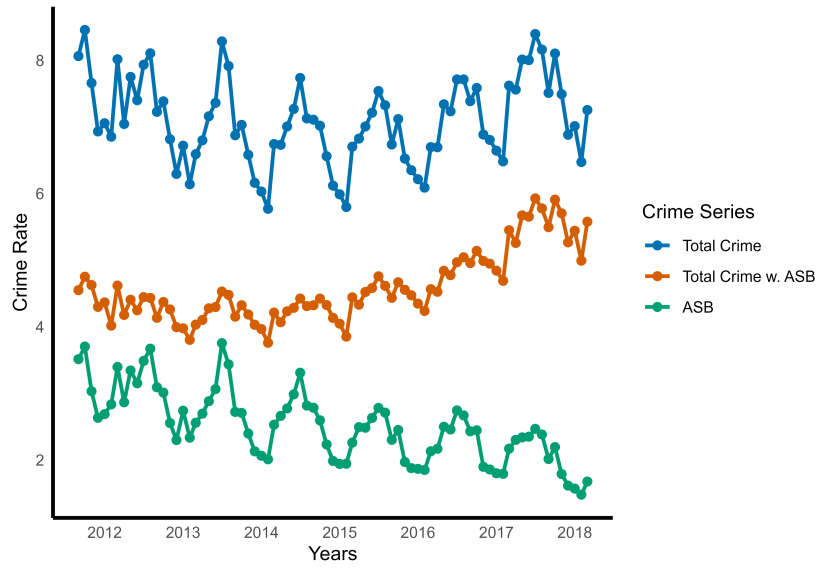
Statistic	N	Mean	St. Dev.	Min	Max
Total Crime	674,265	7.10	7.55	0.00	491.72
Total Crime (exc. ASB)	674,265	4.59	5.57	0.00	393.22
Anti-social Behaviour	920,490	2.45	2.64	0.00	100.00
Violent Crime	674,265	2.18	2.28	0.00	151.24
Property Crime	920,490	2.13	3.34	0.00	293.75
Substance Crime	920,490	0.18	0.50	0.00	121.33

The average monthly crime rate in a ward is roughly 7 instances per 1000 residents. Wards at the 90th percentile have 5.6 times higher crime rates than those at the 10th percentile. The most frequent crime is anti-social behaviour, which accounts for almost a third of all crimes. It is predominantly committed

¹²A summary of the disaggregated crime types is shown in Table A.8 in Appendix A.2. These groups are not exhaustive of all crime offences. A category for ‘all other crimes’ has been left out as it is difficult to interpret as it is an amalgamation of many heterogeneous types of offences. Hence, it is also not included in our ‘total crime’ count.

by young people. For example, out of all Anti-Social Behaviour Orders issued between 1999 and 2013, more than one-third were given to under-18 year olds.¹³ In Figure 3, we report crime trends by type. Anti-social behaviour is highly seasonal and has been declining over the sampled period. This is in contrast to all remaining crimes, which have been rising over the period 2011 to 2019.¹⁴

Figure 3: Crime Rate Time Series (2011-2019)



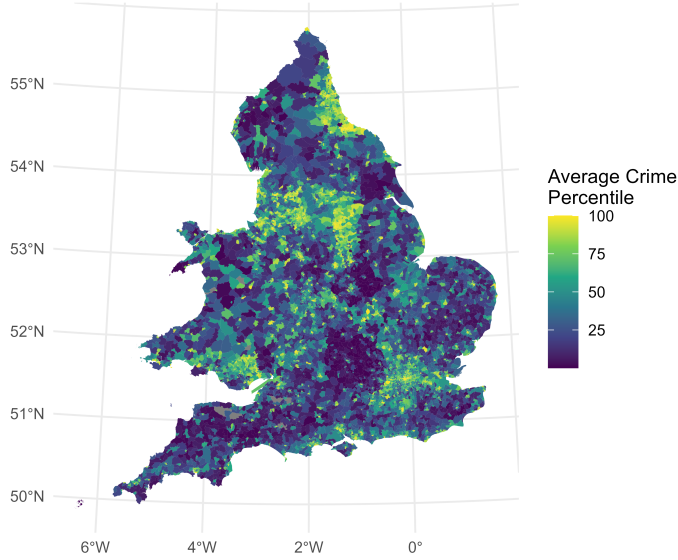
Notes: Values are reported as the average number of incidents per 1000 residents of each ward.

As well as temporal variation, crime displays some location variation, as demonstrated in Figure 4 for ASB. Anti-social behaviour is concentrated in urban areas with most neighbourhoods across England and Wales experiencing low rates of ASB. On average, the 90th percentile neighbourhood experienced 5.6 times more incidents (per 1000 residents) than one at the 10th percentile.

¹³From 1998 to 2014, anti-social behaviour was punishable with Anti-Social Behaviour Order (ASBO) using civil orders, which restricted behaviour or access to locations. From October 2014, ASBOs were replaced with Community Protection Notice, a civil injunction, and Criminal Behaviour Orders, which are issued by magistrate or youth courts. For more details, see Government's Statistical Bulletin on crime.

¹⁴This may appear puzzling as findings from the Crime Survey for England and Wales (CSEW) (explored in the annual statistical bulletin) suggest that such crimes have remained largely stagnant. However, this is partly explained by the scope of the survey which does not consider crimes with no specific victim or those against public bodies.

Figure 4: Location of Anti-Social Behaviour Crime (2011-2019)



Notes: Percentiles are calculated based on the average crime rate for each ward and the total number of projects started over the years 2011-2019. Brighter colours represent a higher percentile.

3.3 Local Area Characteristics

The analysis additionally uses a small set of variables from the 2011 Census for heterogeneity analysis, available at the NOMIS repository. These help to characterise the social and economic characteristics of wards at the beginning of the period of our study approximating the wealth, education and poverty of local areas, as well as their respective degrees of urbanisation, which might affect the level of crime and the probability of receiving lottery funding.¹⁵ In addition, following the 2008/09 financial crisis, the central government implemented austerity measures to reduce governmental debt, characterised by deep cuts to local authorities budgets. Since local authorities are responsible for the provision of social services, including youth services, these cuts might have affected both the crime level (see for example Giulietti and McConnell (2020); Villa (2024)) and the propensity to apply for funding. The austerity cuts are computed as the simulated per person losses (GDP) from the 2012-13 cuts to

¹⁵Table A.9 in Appendix A.3 provides a brief explanation of each variable.

welfare measured at the local authority level, taken from Beatty and Fothergill (2013). The summary statistics on these variables are provided in Table 3.

Table 3: Additional Variable Summary Statistics

Statistic	N	Median	St. Dev.
Population	8,546	5,565	4,961.80
Youth Share	8,546	0.27	0.06
L2 Education Share	8,546	0.48	0.10
Social Rent Share	8,546	0.11	0.11
Urban/Rural	8,546	1	0.47
Austerity Loss	8,365	434	120.92

Notes: This table provides the summary statistics for the key covariates used in the heterogeneity analysis. For set of detailed definitions see Table A.9 in Appendix A.3, it provides a brief explanation of each variable.

4 Empirical Strategy

To estimate the impact of different kinds of projects on crime, we assume that the crime rate evolves according to the causal model in equation (1). This specification allows for heterogeneous treatment effects, and differences in treatment timing across wards for the receipt of small and large projects.

$$y_{it} = \eta_i + \theta_t + \beta_{it}^S D_{it}^S + \beta_{it}^L D_{it}^L + \epsilon_{it} \quad (1)$$

where y_{it} is monthly crime per 1000 persons in ward i at time t , η_i and θ_t are the fixed effects for ward and time respectively, D_{it} is a treatment indicator for the receipt of small or large projects respectively (we discuss other definitions below), and ϵ_{it} is the error term. Note that the duration of treatment differs between project types. For large projects, we assume that once a project is funded, the neighbourhood remains treated for 60 months, i.e. $D_{it}^L = 1$ from start date until 60 months, 0 otherwise. In contrast, we assume that micro projects last at most 24 months, and define the treatment period (i.e. $D_{it}^S = 1$) accordingly. Hence, assuming that no other project begins in the neighbourhood, after the treatment period has elapsed, the ward returns to its untreated state. These assumption reflect funding practice and award rules: In the majority of cases,

large projects are required to last up to 5 years, while micro awards can be used for at most 2 years. From an identification perspective, we require only that treatment effects stabilise by the end of these periods. As discussed in more detail below, we perform several robustness exercises to support the plausibility of these assumptions.

The main concern when estimating the average treatment effect of the funding on crime is that decisions to apply and award funding might be correlated with neighbourhood characteristics which are themselves correlated with crime. Assuming that these neighbourhood characteristics are fixed over time, this endogeneity would be captured by the neighbourhood fixed effects. Hence, we impose that $E[\epsilon_{it} | D_{it}^S, D_{it}^L, \eta_i, \theta_t] = 0$. Combined with the linear functional form of the model, this is equivalent to a typical parallel trends assumption that implies that on average trends in untreated potential outcomes are parallel for all wards.

Conventionally, based on this parallel trends assumption, the average treatment effect on the treated derived from equation (1) could be estimated using a two-way fixed effects (TWFE) model. There are several problems with this approach in our setting. Firstly, the recent work by Goodman-Bacon (2021), Sun and Abraham (2021), Borusyak et al. (2022) and others has shown that under staggered treatment timing and heterogeneous treatment effects the OLS estimator leverages “forbidden comparisons” between ‘already treated’ and ‘not yet treated’ units. Hence, TWFE does not recover the relevant causal effect. Secondly, in designs with multiple treatments, the estimated coefficients will be cross-contaminated by the two treatment effects. Intuitively, if the two treatments are not mean-independent and treatment effects are heterogeneous it becomes difficult to disentangle the separate contributions of each treatment to the evolution of the outcome. In our case, the number of large and micro projects a ward receives is indeed positively correlated (see Figure B.3 in the online appendix).

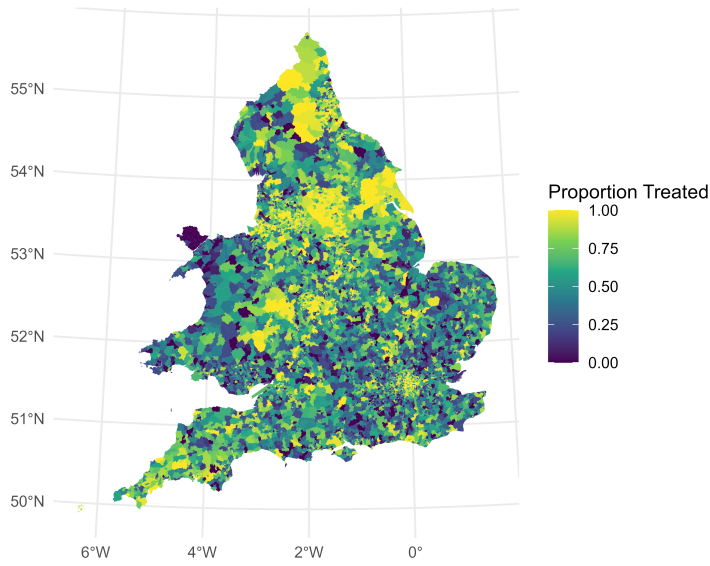
Fortunately, a large number of heterogeneity-robust estimators have been proposed.¹⁶ Due its flexibility in handling different types of treatment variables, we use the two-stage difference-in-differences approach proposed by Gardner (2022). This works in two steps. In the first stage, the time and unit fixed effects are recovered using the sample of untreated observations. It is this step that relies on the parallel trends assumption, which implies that these untreated observations can be used for the imputation. The dependent variables are then “adjusted” by removing the time and unit fixed effects. In the second stage, “adjusted” outcomes are regressed against the treatment variable. This yields numerically identical estimates to those of the “imputation” estimator by Borusyak et al. (2021), though differing in their standard errors. To overcome the treatment effect contamination problem, we use the two-stage difference-in-difference estimator on samples where at most one type of project is active. For instance, when interested in the effect of micro projects, we only work with the sample where $D_{it}^L = 0$.

Intuitively, this estimator relies on exogenous variation in the projects’ start times; i.e. we compare treated wards with those where no projects are currently active. Over the period of analysis, 96% of wards had a project funded at least once (86% had at least one small, and 49% had at least one large project) and on average, neighbourhoods have at least one project active 68% of the time, though with considerable heterogeneity. This variation is well-illustrated in Figure 5, which shows the proportion of time that wards are “treated”. For identification we assume that, conditional on time and ward fixed effects, the start times of projects are orthogonal to trends in potential crime outcomes. This is plausible since the projects do not directly target a reduction in crime and the length it takes to review an application is not known a priori. Treatment

¹⁶See the suggestions in Borusyak et al. (2021); Callaway and Sant’Anna (2021); De Chaisemartin and d’Haultfoeuille (2023); De Chaisemartin and D’Haultfoeuille (2020); Dube et al. (2023); Roth et al. (2023); Wooldridge (2021).

effects of individual projects can then be estimated in a difference-in-difference design with staggered adoption, which compares treated wards to those that have either not yet received a project, or for whom the dynamic effect of a previous treatment has vanished, i.e. they were last treated more than 24 months ago.

Figure 5: Proportion of Time with at least one Active Project (2011-2019)



Notes: Proportions based on the number of periods a ward has at least one active project over the 2011-2019 period. Brighter colours represent a higher proportion.

The identification requires assumptions on unobservable counterfactual trends, which are not directly testable. However, we leverage the periods immediately before the start of any micro project and verify whether pre-treatment trends in crime rates are indeed parallel between treated and to-be-treated wards. In our case, this is a particularly powerful test since the repeat treatment means that we can use pre-treatment periods throughout our sample period. Formally, we follow Borusyak et al. (2021) in validating our design by performing a pre-trend test on the sample of (yet) untreated observations. We estimate the following specification:

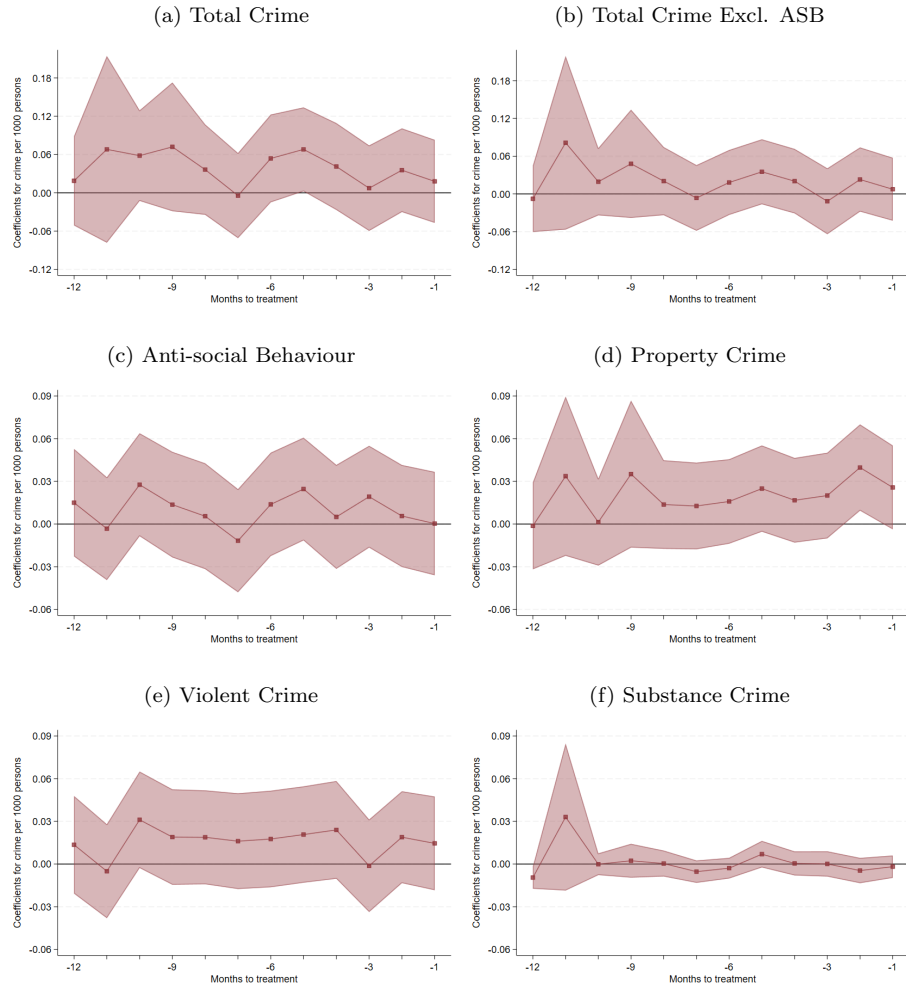
$$y_{it} = \theta_t + \eta_i + \sum_{p=-P}^{-1} \gamma_p 1\{t = \tau_{ij} + p\} + \epsilon_{it} \quad (2)$$

where p denotes the number of months before a unit receives treatment in relative time, τ_{ij} is a treatment date. The test of parallel pre-trend involves an F-test to test of the joint significance of all pre-treatment coefficients (γ). We assume P equals 12 months. The advantage of this approach is that it is external to our choice of estimator, and hence avoids introducing correlations between our estimates and the pre-trend coefficients which, as argued in Roth (2022), can bias results. The results of this exercise are given in Figures 6, each panel illustrating for a type of crime the set of pre-treatment coefficients for the previous 12 months. The associated F-statistic for each regression fails to reject the null hypothesis that all coefficients are jointly zero for all crimes. This supports the assumption that prior to receiving treatment, treated and control wards experienced similar trends in crime.

Having established support for the common trends assumption, we further describe our method for evaluating the effects of small and large projects. Specifically, the term $D_{it}^{S,L}$ in equation (1) is given two interpretations. To start with, we consider the causal effect at the extensive margin: does at least one small/large project reduce crime? Second, since multiple projects can be active simultaneously, we also assess the “dose” response or intensive margin. For micro projects, where more than one can occur at a time, we add a set of dummy variables indicating the level of funding in use. This non-parametrically identifies the effect of having different numbers of projects active in a ward relative to having none at all. Specifically, we define the following categories: less than £1000 (untreated), £1000-£10,000, £10,000-£20,000, £20,001-£30,000, and more than £30,000, roughly equivalent to the ward having 0, 1, 2, 3 or more than 3 projects operating simultaneously, since the median funding for a project is close to £10,000. ¹⁷ It must be noted that without imposing a

¹⁷The interpretation of the estimates are similar if the number of projects is used instead.

Figure 6: Pre-Trend Plots by Type of Crime



Notes: Pre-treatment event study plots estimated using the testing approach suggested in Borusyak et al. (2021). The dependent variable is reported by type of crime per 1,000 residents. A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

strong parallel trends assumption that holds across all ‘dosage’ states (intensive margin), we cannot identify the marginal effect on the crime rate of increasing funding from say £20,000 to £30,000, but only the difference between these dosages and having no projects active at all Callaway et al. (2024). For large projects, we instead create six bins: £0 (untreated), £1-£100,000, £100,001-200,000, £200,001-300,000, £300,001-400,000, £400,001-500,000, and >£500,000 that match the size of funding more appropriately.

Next, for both the extensive margin and dose response, we perform heterogeneity analysis by 1) urban versus rural status, 2) socio-economic indicators and 3) type of projects. Additionally, Giuliatti and McConnell (2020) found that areas most affected by the 2012 UK Welfare Reform Act had increased crime level during the austerity period. We thus assess whether the impact of micro-project differs in neighbourhoods more affected by the austerity policies and estimate the following model (3):

$$y_{it} = \eta_i + \theta_t + \beta_{it}D_{it} + \gamma_{it}Aust_{it} + \tau_{it}(Aust_{it} \times D_{it}) + \epsilon_{it} \quad (3)$$

where the variables have the same interpretation as above, but $Aust_{it}$ is a treatment indicator for whether a ward is above median in term of budget loss due to the austerity cuts starting in April 2013, as calculated by Beatty and Fothergill (2013). As above estimation proceeds using the two-stage difference-in-differences approach. Importantly, our comparison group is the set of ward-period combination that neither have an active project nor are highly affected by austerity in an approach that resembles a triple difference-in-difference design.

Finally, we conduct a battery of robustness checks. First, we assess whether local interventions reduce crime or simply displace it to other neighbourhoods. We test for the presence of spatial spillover effects by including, in the second

For the full set of results of this, see Table B.1 in Appendix B.

stage regression, dummy variables for whether a neighbouring ward is in receipt of a small or large project.¹⁸ Second, we evaluate the sensitivity of the main results to changes in the sample restrictions: we check whether our baseline results depend on the length of the activity window of small (18, 24, or 30 months) and large (36, 60, or 84 months) projects, as well as type of projects considered (by including non-youth projects in the sample).

5 Results

5.1 Micro-projects

Table 4 shows the extensive and intensive margin two-stage difference-in-difference (DiD) estimates of exposure of a local area to active micro-projects. For the extensive margin (Panel A), we compare changes in crime rates in areas with at least one micro project to those without an active project (the not-yet treated) over the same time period. In addition, we investigate the dose-response effect along the intensive margin (Panel B) to investigate whether increased investment in a ward, i.e funding more (small) projects, has a cumulative effects on crime. In doing so, we compare areas that receive different quantities of funding to those that are in receipt of no funding. As mentioned above, we measure funding increases by multiples of £10,000, which approximately corresponds to one additional micro project.¹⁹

¹⁸Note that this is only a test of whether the presence of a treatment effect in a given ward can be explained by the treatment status of a neighbouring ward. A major shortcoming of this approach is that we cannot model any spatial contamination of control wards. This is because there are few periods when both a given ward and its neighbouring ward are untreated at the same time.

¹⁹Note that the estimates become less precise as the amount of funding in a ward increases since the number of periods where wards receive that many project is fewer.

Table 4: Micro-projects and Local Crime

	(1) Tot.	(2) Tot. Excl. ASB	(3) ASB	(4) Prop.	(5) Viol.	(6) Subst.
Panel A: Extensive Margin						
Micro	0.002 (0.017)	0.018 (0.013)	-0.028*** (0.010)	0.020*** (0.007)	0.008 (0.009)	0.001 (0.002)
Panel B: Intensive Margin						
Up to £10k	-0.002 (0.015)	0.007 (0.012)	-0.011 (0.010)	0.011* (0.006)	-0.004 (0.008)	0.001 (0.002)
£10k-£20k	-0.005 (0.036)	0.034 (0.024)	-0.051*** (0.019)	0.033*** (0.012)	0.024 (0.016)	-0.000 (0.003)
£20k-£30k	0.024 (0.054)	0.038 (0.046)	-0.079*** (0.030)	0.036* (0.021)	0.030 (0.033)	-0.004 (0.005)
£30k+	0.133 (0.103)	0.156* (0.085)	-0.142** (0.057)	0.081* (0.044)	0.166*** (0.062)	0.000 (0.007)
Mean	7.167	4.635	2.474	2.160	2.199	0.179
Rural	4.124	2.690	1.402	1.312	1.231	0.093
Urban	8.639	5.577	2.994	2.571	2.668	0.221
Percentile	51.363	52.372	46.317	52.466	54.507	54.851
N	353765	353765	491646	491646	353765	491646

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the two-stage difference-in-difference estimation results showing the average treatment effect of receiving micro-projects on the monthly crime rate at the ward-level. Panel A shows the estimates for the effect of receiving at least one micro-project, while Panel B has the estimates for the effect of receiving different numbers of micro-projects concurrently in £10,000s. The crime rates running from models (1) to (6) are total crime, total crime excluding anti-social behaviour (ASB), anti-social behaviour (ASB), property crime, violent crime and substance crime respectively. The mean values for each type of crime, and their values in rural and urban wards are also reported alongside the change in the percentile of the crime distribution that a ward would see given the estimated treatment effect. Standard errors are given in parentheses and are clustered at the ward-level.

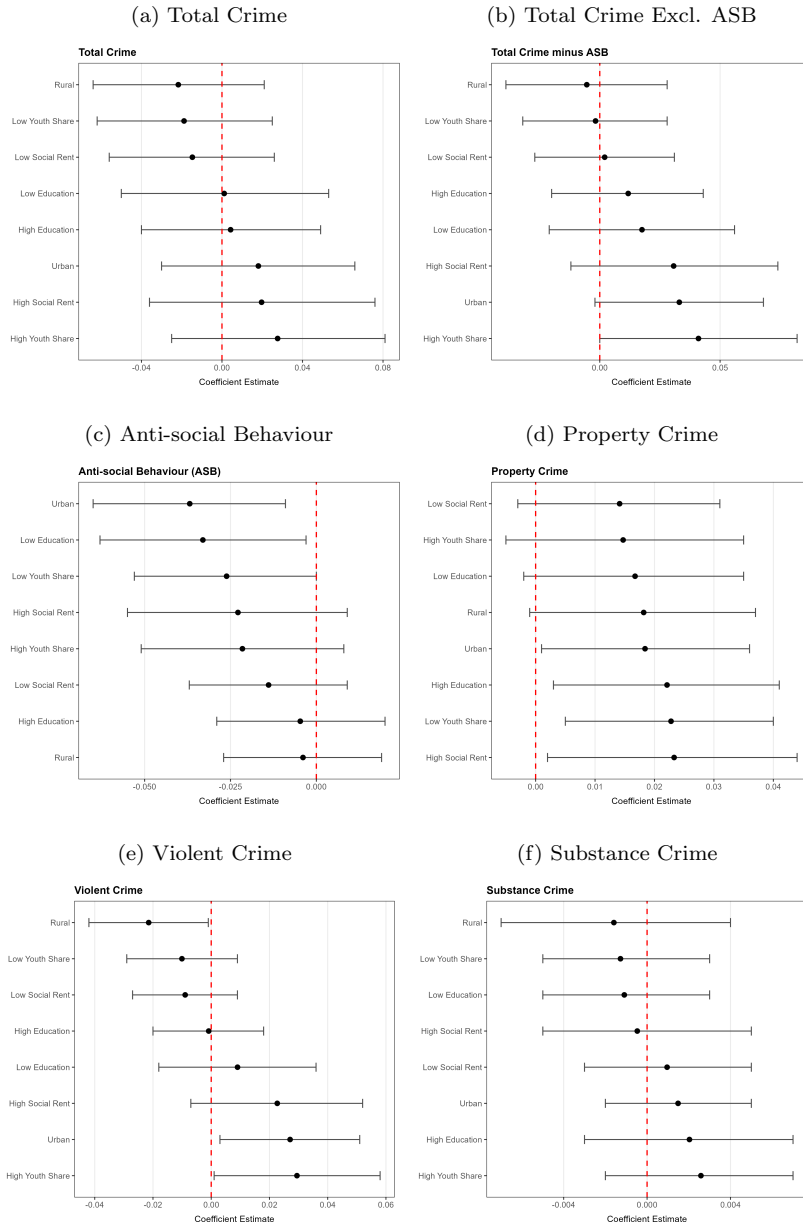
In Panel A, while we find no statistically significant net effect on overall crime, we estimate an important shift in crime activity: receipt of at least one project reduces anti-social behaviour by just over 1% and increases property crime by just under 1% (relative to the monthly mean).²⁰

Though small investments of up to £10,000 worth of projects display few detectable effects, as shown in Panel B, investments also do not appear to affect overall crime, when multiple micro-projects are simultaneously active. However, we find a similar shift in crime types as at the extensive margin. Increased investment leads to greater reduction in reported anti-social behaviour, which drops by nearly 6% when more than three micro-projects are running simul-

²⁰Event study plots for the effect of micro project on each type of crime are presented in Figure B.5 in Appendix B. Consistent with the ATT estimates, there are few statistically significant effects.

taneously, with monotonously rising impacts as investment increases. This is counteracted by a (similarly monotonous) rise in property and violent crime by 3.7% and 7.5% respectively.

Figure 7: Heterogeneity Plots by Type of Crime

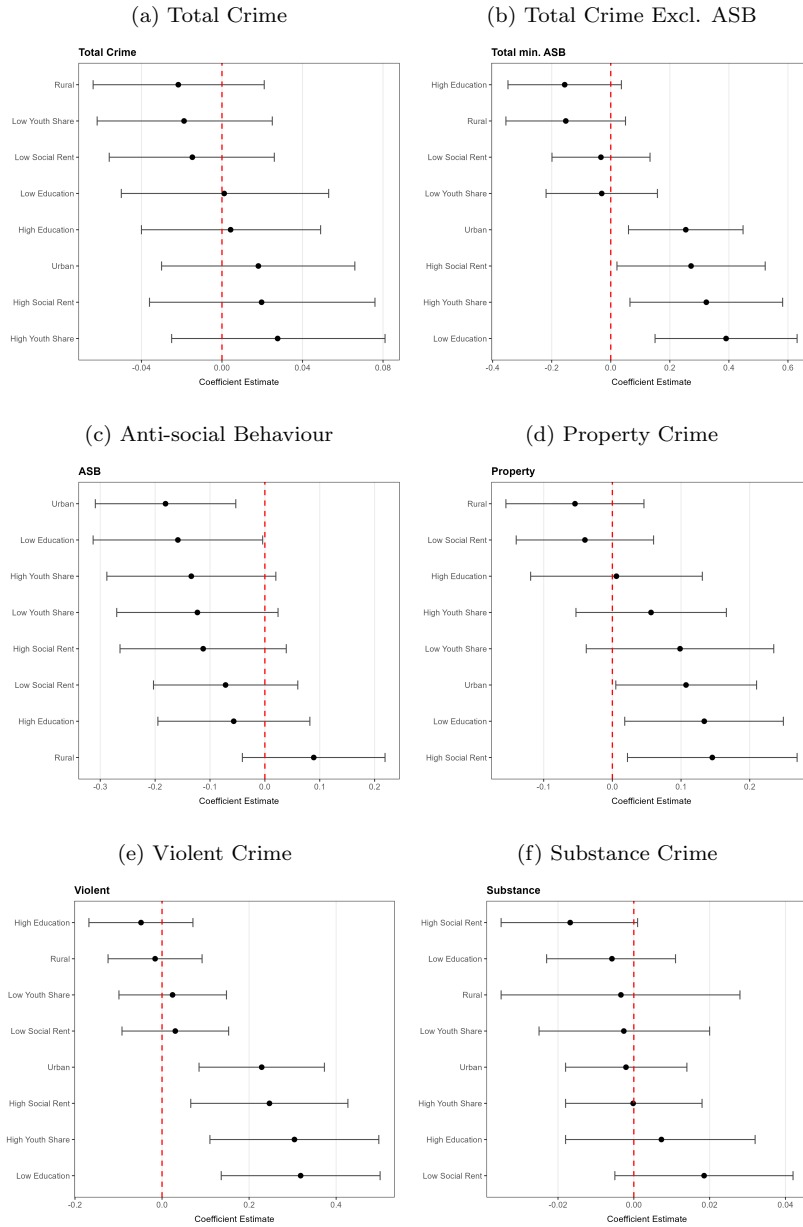


Notes: A set of heterogeneity plots for the effect of receiving at least one micro project on different types of crime. The dependent variable is reported by type of crime per 1,000 residents. A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

Next, we show that these results mask considerable heterogeneity across ward characteristics. We re-estimate the model on the extensive margin but split the

sample of areas into the top or bottom quartiles of the distribution of a particular observable. Figure 7 plots the estimates for each type of crime. A basic pattern emerges: urban areas, areas with a large share of young people and - to some extent - those with a higher fraction of residents with lower levels of education see stronger negative effects on anti-social behaviour (ASB), but increases in property and violent crime. We find no detectable impact on substance crime.

Figure 8: Heterogeneity Plots by Type of Crime



Notes: A set of heterogeneity plots for the effect of receiving at over £30,000 worth of funding for micro projects on different types of crime. The dependent variable is reported by type of crime per 1,000 residents. A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

Returning once more to the intensive margin, Figure 8, which presents the estimates for receiving £30,000+ worth of funding for micro projects split by

various ward-level characteristics, shows similar concentrations of impacts in urban areas, areas with lower levels of education and a high share of young residents as shown previously when considering the extensive margin.²¹

Anti-social behaviour falls most in these areas (around 6-7%), but, conversely, property and violent crime rises the most, by around 5-6% and 10-13% respectively. The explanation for this pattern, similar to Jacob and Lefgren (2003), likely lies in the interplay between the “concentration” effect and the “incapacitation effect”. Although many micro projects keep vulnerable youths off the streets and busy, the provision of many social activities also runs the risk of creating conflict between youths that could become violent. In addition, providing potentially crime-prone young people with more contacts from similar backgrounds allows for the exchange of information, including on new crime targets.

One of the advantages of our approach is that we can estimate the effectiveness of different types of projects on crime, as summarised in Table 1. We use the quantity of funds going towards projects of various types (in £10,000s) as the treatment variable. This approach is valid under the assumption that the ‘dose’ curve is linear and operating on the intensive margin, evidenced by the results shown in Table 4.²² The full sets of results are given in Tables B.3 and B.4 in Appendix B. The results for the extensive margin estimates are very similar to the ones for the intensive margin largely because it is rare for projects of the same type to be active in the same place at once.

We find that an extra £10,000 reduces anti-social behaviour by just under 1% when used for Health & Well-being projects, but raises both property and

²¹The difference that splitting by urban/rural wards makes is illustrated well by Figure B.4 in Appendix B.

²²This is done simply to preserve a sufficiently large sample size for each project type, but may be subject to cross-contamination because projects of different types can be active in the same ward at a time. However, excluding these situations reduces the sample size for the rarer project types. Hence, we do not claim that these estimates necessarily have a causal interpretation, but are nonetheless suggestive of what is most effective.

violent crime by roughly the same amount. This is evidently a compositional change in crime. Many projects belonging to this category, which accounts for 58% of all projects, are sporting events, after-schools clubs and similar activities that may create an incapacitation effect by keeping youths busy, thus reducing ASB, but creating a concentration effect by bringing them together (increasing property and violent crime). This second effect is also very strong for the smaller fraction of projects in the Civic and Social Engagement category, causing an increase in violent crime by almost 6%. This category, making up 2% of all projects, comprises projects that, for example, involve campaigning and protesting for political/social change, thus creating opportunities for confrontations and volatile interactions between large groups of young people.

More positively, ASB is significantly reduced by almost 2% by Skills and Opportunities projects, which account for 17% of all projects. The coefficient given here is likely an under-estimate because such projects would have life-time effects that often cannot be detected over our sample period. The complementary second classification divides projects into those focused on Investment & Infrastructure, such as new sports equipment, or providing activities & services.²³ Impacts are concentrated among projects that provide Activities and Services. ASB is reduced by around 1.5%, but in parallel property crime increases by 1%.

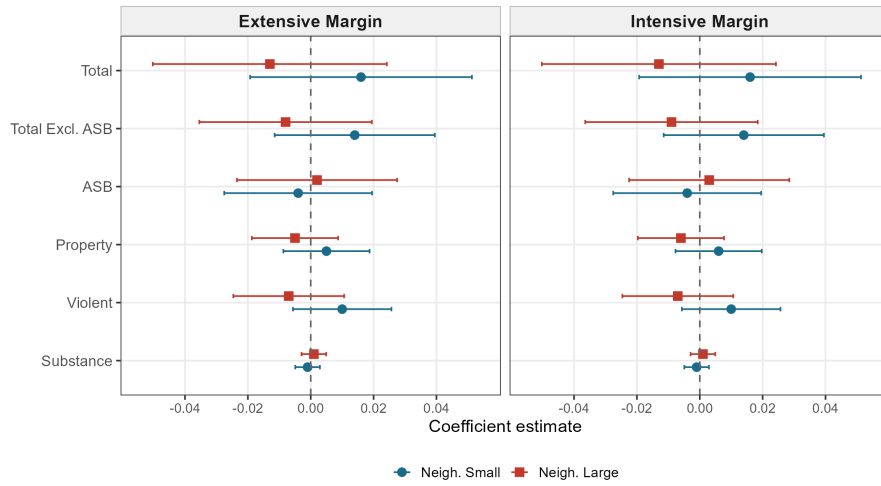
Overall, heterogeneity across projects reveals that those focused on supporting the physical and emotional well-being of youths can reduce the number of instances of anti-social behaviour in the local area - but at the cost of rising property and violent crime. In contrast, educational projects focused on Skills & Opportunities reduce ASB without counteracting impacts on other crimes.

²³The category Other comprises projects where this delineation is not clearly evident from the project description.

5.1.1 Robustness Checks

First, we test for the presence of spatial spillover effects. This is done by seeing whether any of the difference between the actual outcomes and those predicted by the first-stage of our difference-in-difference model are associated with the treatment status of neighbouring wards. The estimates for both the extensive and intensive margin are presented in Figure (9). Across all models, the coefficients for both the presence of small and large projects are not significant. This suggests that neighbourhood spillover are likely not drivers of our results.

Figure 9: Spatial Spillover Effects and Micro Projects

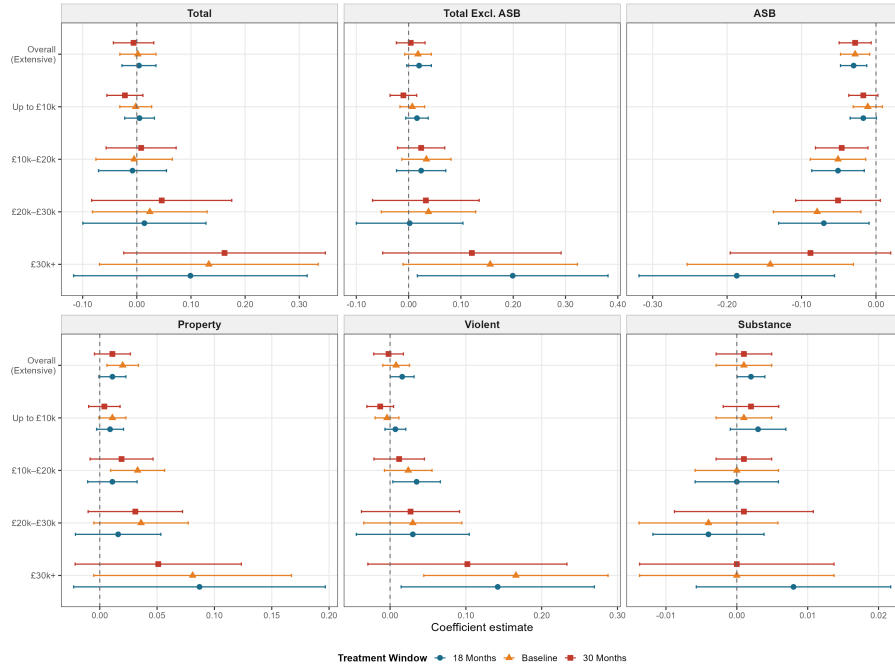


Notes: A plot of treatment effects estimates for the impact of a neighbouring LSOA receiving funding for micro projects on crime. A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

Second, we check whether effect change when the activity window for micro projects is altered to either 18 or 30 months. Large difference from the baseline specification would indicate that either the dynamic effects of receiving funding is concentrated closer to the start date, or that the impact is considerably longer-lasting than initially assumed. Figure 10 reports the different estimates from our extensive and intensive margin specifications. Visually, the estimates are strongly consistent with one another and there are no statistically significant

different between them. In several cases, the 18 month estimates are larger in magnitude suggesting that when the activity window is extended to 30 months, the model captures many null effects, bringing down the estimated treatment effect. Nonetheless, the results from these models are consistent with our baseline estimates.

Figure 10: Micro Project Estimates by Activity Window

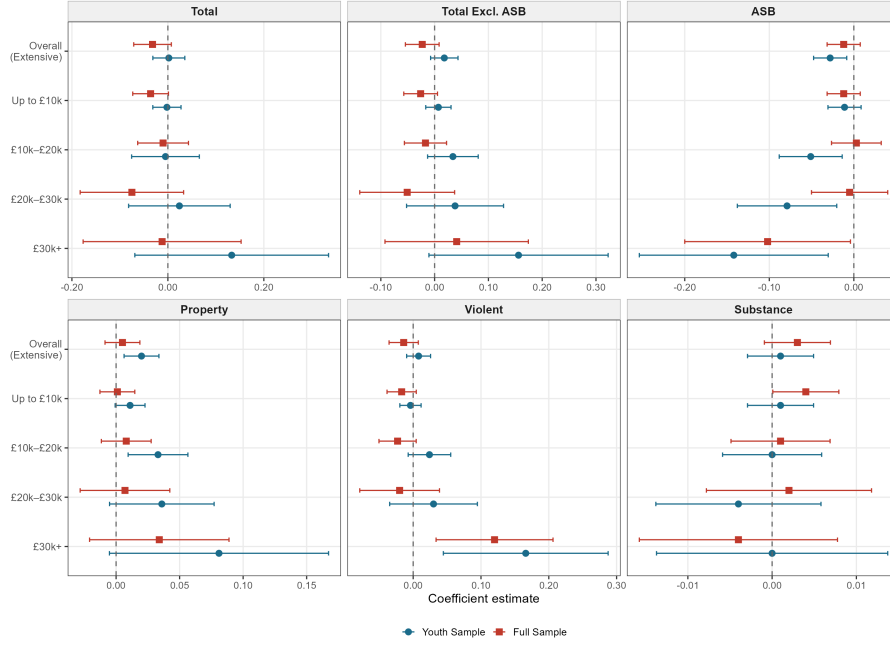


Notes: A plot of treatment effects estimates for the impact of receiving different quantities of funds for micro projects on crime by whether an 18-, 24-, or 30-month activity window for micro projects is used. A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

Finally, we show that our baseline results are robust to the choice of sample. As described above, our focus is on youth-related projects and so projects targetting other groups are excluded as part of our set of sample restrictions. Figure 11 illustrates our extensive and intensive margin estimates across both samples. In each case, the youth-sample estimates are usually larger in size than those of the full sample. This is not surprising as the full sample estimates are absorbing the impact of projects targetting the elderly or very young children that would not be relevant to the local crime rate. Overall, though estimates on the full

sample are visibly more muted, they remain broadly consistent with our baseline results.

Figure 11: Micro Project Estimates by Project Sample



Notes: A plot of treatment effects estimates for the impact of receiving different quantities of funds for micro projects on crime split by whether the model is estimated on the full or youth-project samples. A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

5.2 Large Projects

In our main specification estimating the impact of large projects on crime, we only focus on those neighbourhood-time pairs with no active micro projects. This allows us to estimate both the impact of receiving at least one large project, and to investigate whether the quantity of funding makes a difference. Table 5 presents the results along both margins. Starting with the extensive margin in Panel A, we find that in contrast to micro-projects, there is no effect on anti-social behaviour. Moreover, total crime excluding anti-social behaviour declines by about 1.6%. This can be attributed to a decline in violent crime that falls by nearly 4%, but is partly offset by the 1.3% increase in property crime.

Table 5: Extensive Margin for Large Projects

	(1) Tot.	(2) Tot. Excl. ASB	(3) ASB	(4) Property	(5) Viol.	(6) Subst.
Panel A: Extensive Margin						
Large	-0.064* (0.037)	-0.073*** (0.028)	0.018 (0.022)	0.029** (0.014)	-0.085*** (0.019)	0.002 (0.003)
Panel B: Intensive Margin						
<£100k	-0.032 (0.044)	-0.048 (0.034)	0.024 (0.027)	0.034* (0.018)	-0.059*** (0.023)	0.002 (0.004)
£100k-£200k	-0.044 (0.101)	0.016 (0.077)	0.001 (0.058)	-0.006 (0.033)	0.004 (0.051)	0.001 (0.007)
£200k-£300k	-0.288* (0.174)	-0.233** (0.114)	-0.007 (0.083)	-0.003 (0.051)	-0.136 (0.093)	-0.017** (0.008)
£300k-£400k	0.068 (0.213)	0.094 (0.145)	0.016 (0.104)	0.141** (0.061)	-0.040 (0.100)	0.002 (0.016)
£400k-£500k	-0.218 (0.164)	0.035 (0.143)	-0.335** (0.132)	0.131** (0.061)	-0.060 (0.103)	0.036 (0.025)
£500k+	-0.203 (0.197)	-0.223 (0.147)	0.068 (0.101)	-0.018 (0.060)	-0.103 (0.114)	-0.012 (0.022)
Mean	7.167	4.635	2.474	2.160	2.199	0.179
Rural	4.124	2.690	1.402	1.312	1.231	0.093
Urban	8.639	5.577	2.994	2.571	2.668	0.221
N	266336	266336	380335	380335	266336	380335

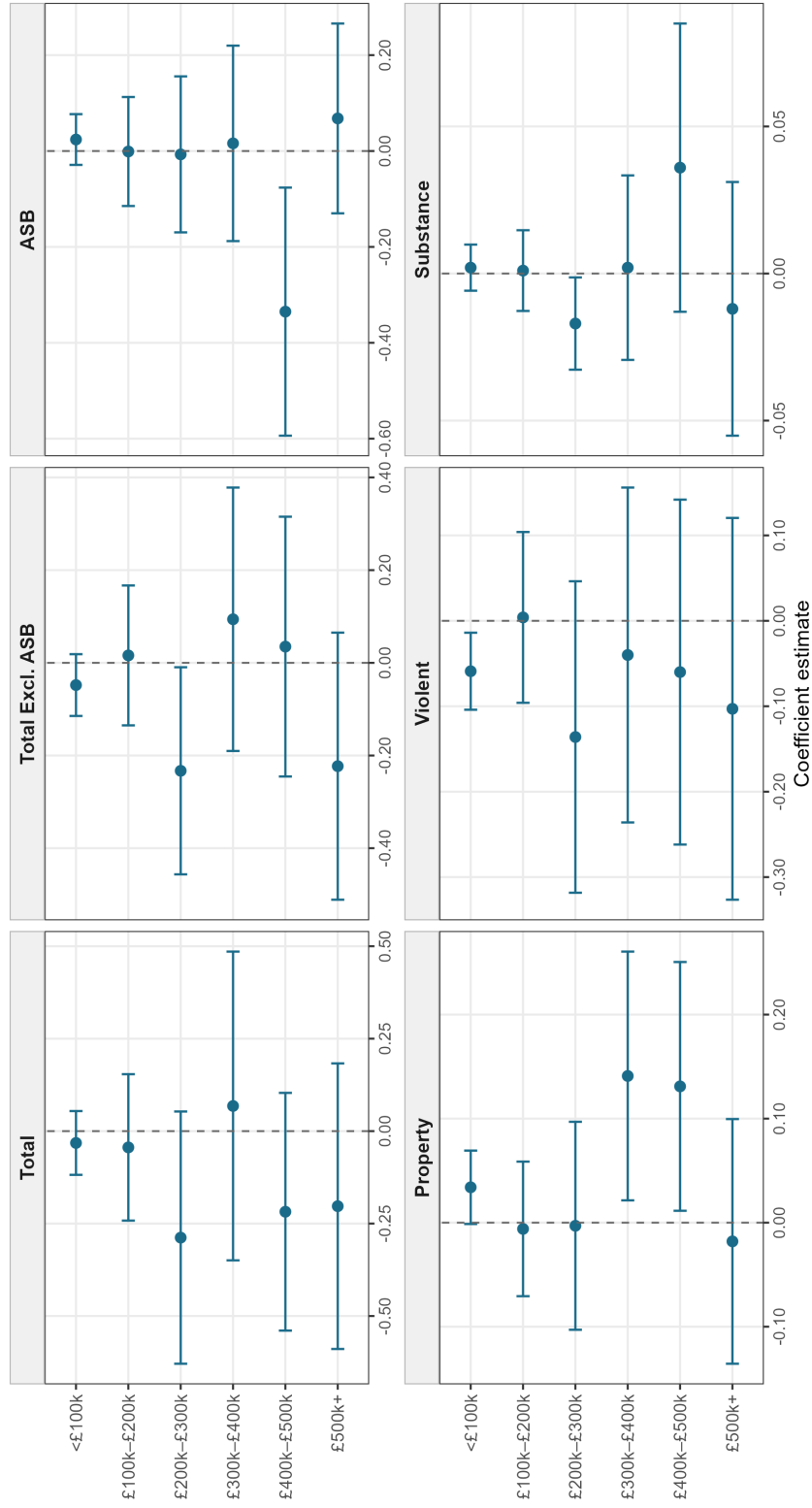
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the two-stage difference-in-difference estimation results showing the average treatment effect of receiving large projects on the monthly crime rate at the ward-level. Panel A records the results along the extensive margin, i.e. receiving at least one large project. Panel B shows the results along the intensive margin reflecting the impact of projects of different based on their respective funding amount. The crime rates running from models (1) to (6) are total crime, total crime excluding anti-social behaviour (ASB), anti-social behaviour (ASB), property crime, violent crime and substance crime respectively. The mean values for each type of crime, and their values in rural and urban wards are also reported alongside the change in the percentile of the crime distribution that a ward would see given the estimated treatment effect. Standard errors are given in parentheses and are clustered at the ward-level.

To understand whether the quantity of allocated funds matters, we estimate the effect of receiving a large project by size category. In this case, the results only have a monetary interpretation as it is exceedingly rare for there to be more than one large project active in a given neighbourhood at a time. Hence, we can see what investments belonging to the bins: £1-£100,000, £100,001-200,000, £200,001-300,000, £300,001-400,000, £400,001-500,000, and >£500,000 have the largest impact. These estimates are illustrated in Figure 12. The crucial detail in this diagram is that there are no systemic patterns for any time of crime. That is, there are no cases for which greater funding leads to a monotonic change in crime. More concretely, although for ASB, projects in the £400k-500k range appear most effective, the causality is backwards: it is not the greater quantity that is driving the effect, rather the project that was effective just so happened

to belong to that range. This suggests that efficacy of projects cannot be in there mere size, but rather in how optimally they meet the needs and context of the local community. For this reason, we focus on the extensive margin for the rest of the results.

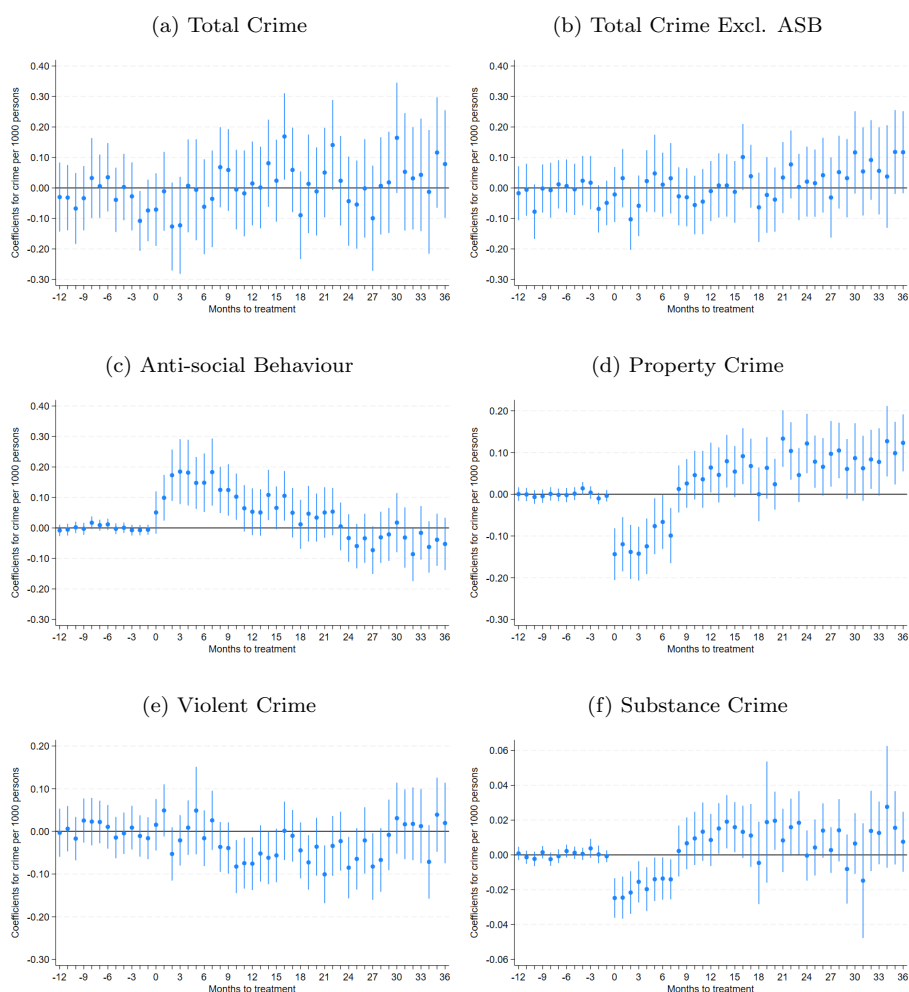
Figure 12: Intensive Margin Effects of Large Projects



Notes: A plot of treatment effects estimates for the impact of receiving different quantities of funds for large projects on crime. A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

These static effects hide considerable heterogeneity across time. To see this consider Figure 13 which presents the event study plots for receiving a single large project. Though there may not be much of an effect on anti-social behaviour in the static model, the dynamic effects imply that it rises by nearly 10% in the first few months of the project. On the other hand, both property and substance crime see strong reductions of up to 7% and 12% respectively, lasting about 7 months. Subsequently property crime recovers and begins to grow, while the effect on substance crime disappears. Violent crime also sees a perceptible fall in the first 30 months reaching a decline of just under 5%. These findings imply that policymakers need to distinguish sharply between the short-run and long-run effects of these larger community projects. Although they provide more opportunities for vandalism and theft, these larger investments are able to redirect the energy of more vulnerable young people, leading to a lower incidence of violence.

Figure 13: Event Study for Large Project

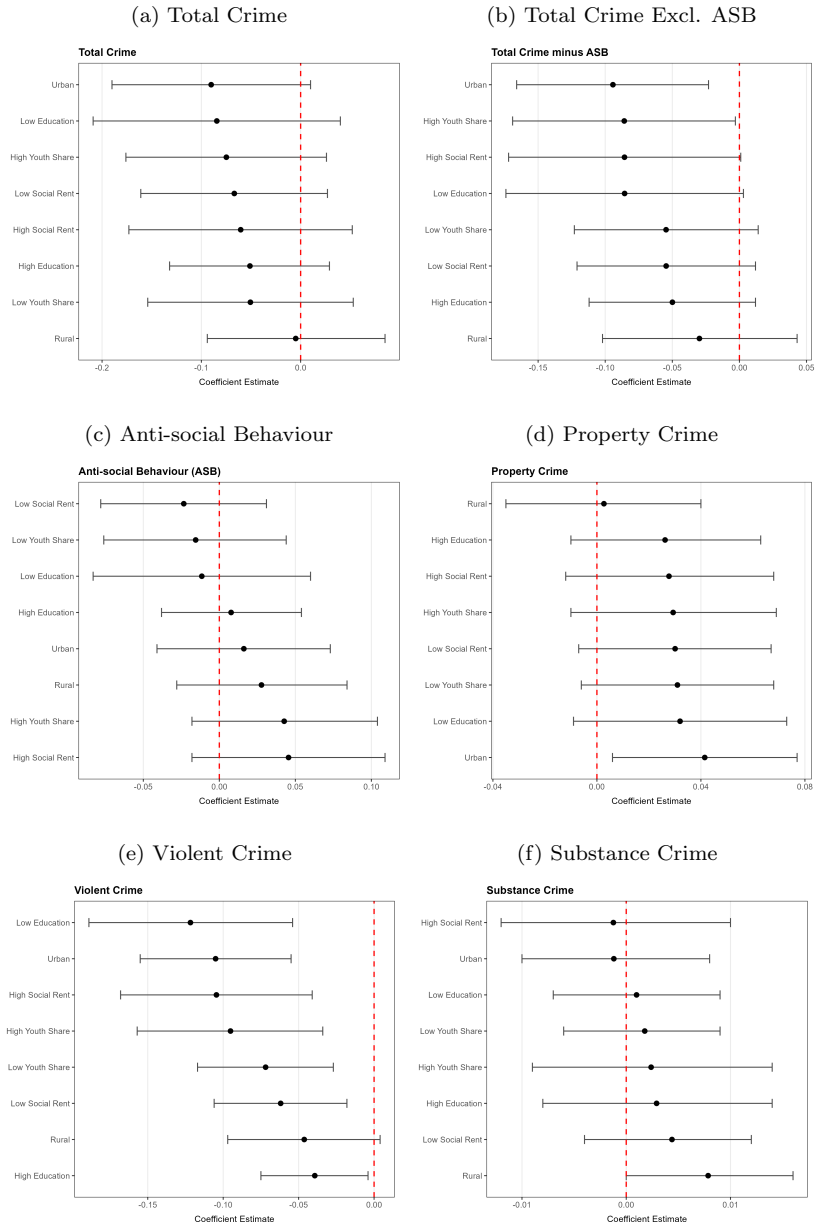


Notes: A set of event study plots for the effect of receiving a large project on different types of crime. The dependent variable is reported by type of crime per 1,000 residents. A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

Certain areas may be more receptive to these large local projects than others, depending on their socio-economic conditions. Here the patterns are consistent with what was seen for smaller projects as effectiveness is greatest in urban neighbourhoods with a high share of young residents (aged 0-24) and a large share of residents living in socially rented accommodation. This is particularly true for violent crime that falls nearly 6% in wards with a low average level of

education. Although the estimates are no significant, ASB appears to rise in those neighbourhoods. This provides some limited evidence that the displacement effect seen in smaller projects is reversed for larger ones. For the small interventions, ASB fell but both property crime and violent crime rose in response, whereas for the larger project ASB and property crime rise, but violent crime declines. This appears to suggest that no matter what type of project one organises, it will invariably provide greater opportunity for thefts. However, whereas larger interventions can curtail some of the more passionate impulses that often characterise interactions between youths, smaller interventions are insufficiently prolonged to have the same effect. One can illustrate this intuition using a simple example. Compare organising a set of sports events with the construction of a new sports centre. Both cases provide opportunities to both steal items and for thieves to exchange information and increase their networks, hence leading to a rise in property crime. However, the sports days keeps youths busy reducing ASB, but the competition may lead to tension and potentially violence. The construction of the sports centre, on the other hand, may lead to vandalism or simply disruptive behaviour outside the building but would provide more vulnerable youths with an opportunity to channel their passion towards meeting their sporting goals.

Figure 14: Heterogeneity Plots by Type of Crime



Notes: A set of heterogeneity plots for the effect of receiving a large project on different types of crime. The dependent variable is reported by type of crime per 1,000 residents. A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

As the quantity of funds appears irrelevant to the effectiveness of large projects, it may be that the type and objectives of the interventions may be of greater

importance. In this way, it may be that projects of a specific type are better at addressing the problem of the local community. Table B.5 in the Online B present the estimates across both project classifications. We find that ‘Health & Wellbeing’ projects both increase property and reduce violent crime by about 1.5% and just under 2% respectively. In both cases, the estimates are only weakly significant. Interestingly, projects focusing on ‘Skills & Opportunities’ reduces violent crime by about 7%. This could represent the positive results of youths, particularly those from households closer to the bottom of the income distribution, being provided with new education and employment opportunities. Alternatively, these could be simply incapacitation effects as in these workshops there are fewer opportunities for volatile interactions that could arise in a competitive setting. The results for the second are less conclusive: we find that ‘Activities & Services’ projects reduce crime excluding anti-social behaviour by just under 3% though it is not possible to pinpoint which specific type of crime this result is driven by. On the other hand, consistent with the intuition described above, the same kind of projects increase property crime by roughly the same percentage.

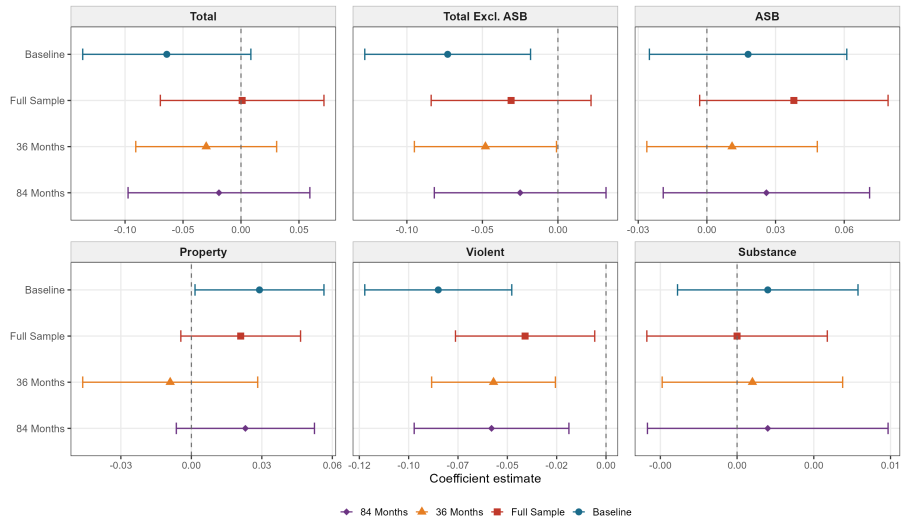
5.2.1 Robustness Checks

The robustness of the large project results is evaluated along similar dimensions to the micro projects. As before, we first consider whether our results can driven by spatial spillover effects. This is a stronger possibility for larger projects that are often much larger and hence whose effects may be more geographical diffuse. Hence, as before, we add indicators for the treatment status of neighbourhood wards as additional explanatory variables in the second stage regression. The results of this are presented in Table B.6 given in the Online Appendix. In each case, the spillover variables are not significant.

Furthermore, we consider whether our estimates are sensitive to our sample restrictions and to the assumed activity window of the larger projects. In par-

ticular, once more, we rerun our baseline model for the full sample that includes all projects (not just those classed as ‘youth’ projects), as well as for the case when large projects are assumed to be active for either 36 or 84 months. The results of these exercises are given in Figure 5.2.1 below.

Figure 15: Large Project Estimates by Activity Window and Sample



Notes: A plot of treatment effects estimates for the impact of receiving different quantities of funds for micro projects on crime by whether a ward is affected by austerity policies (cuts are below or above the median). A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

As expected, the full sample estimates are usually biased in the zero direction across all specifications. Nonetheless, they mostly take the same sign as our baseline results. Altering the activity window has a similar effect. When estimates are restricted to just 36 months, many of the long-run effects are not captured, while extending to 80 months usually includes too many null effects and restricts the sample of ‘control’ periods leading to less efficient estimates. Nonetheless, the key insights remain: large projects act to reduce violent crime, have a largely positive effect on anti-social behaviour and property crime.

6 Conclusion

In this paper, we present the first causal estimates of the effect of grassroots-led investments on local crime rates in England and Wales over the 2011-2019 period. We rely on innovative data from the National Lottery Community Fund to construct a large panel of community projects targeting young people, and combine it with monthly crime rates at the electoral ward-level from the Police Reporting Archive. We exploit differences in the timing of these projects which provide a unique opportunity to study the impact of place-based policies on crime.

We find no detectable effects of micro projects on overall crime. In contrast, our evidence shows that such place-based funding of micro projects lead to compositional changes in the crime. More serious offences such as property and violent crimes tend to rise while lower level crimes, such as anti-social behaviour, drop. Interventions costing as little as £10,000 increase property crime by just under 1% over a two-year period. A greater number of concurrent projects of £30,000 or more leads to larger compositional changes - with property crime increasing by 3.7%, and violent crime by almost 7.5%. The drop in anti-social behaviour of just under 6% only partially compensates for this. These compositional changes might be due to agglomeration effects, whereby projects provide a focus point for criminal activities. Unsurprisingly, these impacts are larger in urban areas with higher shares of younger residents, as well as those with lower levels of education and with higher shares of those living in socially rented accommodation. Investment into larger projects, in contrast, do lead to a reduction in overall crime by just under 1%. Additionally, they lead to a long-run decline in more serious crimes, i.e. violent crime falls by 4%, with a counteracting, modest increase in property crime of about 1.6%.

The diverse nature of funded projects offers the opportunity to better under-

stand which investments are particularly beneficial: small and large investments into projects oriented at improving ‘Skills & Opportunities’ reduce crime unequivocally, with anti-social behaviour dropping by nearly 2% and violent crime by 7%. These estimates are consistent with the large body of work showing the returns to education as a crime-deterrent strategy. They also imply that tailoring this funding towards investments that enhance skills and opportunities for young people could make them a helpful tool to reduce youth crime.

Our results suggest that untargeted grass-root-led interventions have only limited potential to prevent crime. Micro projects mostly reshuffle it from (lesser) anti-social behaviour crimes to more serious property and violent crime, and large-scale funding reduces the number of violent incidents at the cost of greater property crime. Yet, where funding is directed towards enhancing skills and opportunities, the picture changes: these interventions have the potential to enact declines in overall crime by strongly reducing both ASB and violent crime. Thus, we contribute to the debate on optimal public policy responses to crime.

Although few studies have discussed the efficacy of similar interventions, our estimates are of a similar magnitude to those evaluating the various after-effects of austerity. For instance, Giulietti and McConnell (2020) find that districts exposed to welfare reforms - which resulted to drop in public services, see a 3.5% increase in total crime and a 3.9% rise in violent crime. Equally Villa (2024), finds that as a result of youth centre closures in London, juveniles committed 15% more crimes. That being said, we also cannot recommend either small or larger projects as substitutes for these public services, as crime often rises following a new project in areas strongly affected by austerity cuts.

Overall, the implication for public policy is that community-led intentions, if targetted and chosen appropriately, can be cost effective at reducing some type of crime. Moreover, projects that focus on young people with the aim of helping with career aspirations and providing sporting opportunities may be an effective

means of reducing anti-social behaviour and disrupting behavioural patterns that lead to more serious forms of crime.

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A Data Appendix

In this *Data Appendix*, we detail the process by which the crime and funded project datasets are constructed, as well as any underlying problems in the data.

A.1 Funded Project Data

A.1.1 Sources

Data for funded social projects is taken from two major distributors: the National Lottery Community Fund and Sport England. The former makes up the majority of projects, but includes funding for trusts that themselves redistribute funds nationwide. The largest of these are the (Big) Local Trust and the Power to Change Trust. All data is publicly available and links are provided in Table A.1 below.

Table A.1: Raw Funded Project Data Sources

Name	Link
National Lottery Community Fund	Source
Sport England	Source
(Big) Local Trust	Source
Power to Change Trust	Source

Notes: This table reports the distributor for the funded projects and the link to where project data can be accessed.

A.1.2 Data Cleaning

The raw data from all four sources yields a sample of 292,717 projects: 249,488 by the National Lottery Community Fund; 40,258 by Sport England; 2,821 by the Power to Change Trust, and 150 by the Big Local Trust. Initially, we remove all projects not taking place in England and Wales, and subsequently proceed to matching each project to the respective ward in which it was organised. The raw data collected by the National Lottery Community Fund for projects undertaken across England and Wales detailed 160,517 funded projects containing

any ward-level geographic identifiers. However, in some entries, this information was erroneous and a geographic matching procedure was undertaken to recover the ward-level location of these projects. The Sport England, Big Local and Power to Change Trust data contained the exact co-ordinates for the project beneficiary, and hence no matching procedure was required. Instead, Full Extent Boundary shape-files were used to construct a one-to-one matching between the co-ordinates and ward code. As a result, out of the 160,517 National Lottery Community Fund projects, the locations of 157,902 were identified, corresponding to a 98% success rate. The matching procedure itself followed a number of steps:

1. Using a look-up table from 2011 Wards to 2011 Census Merged Wards, matching was performed on the basis of ward names and local authority names (to avoid duplicates)
2. A partial match function was used to identify ward codes for those wards that had undergone slight name changes or whose name were inputted with error while being in the same local authority
3. An (imperfect) correspondence of 2011 wards was created with ward codes from 2015, 2016, 2017, 2018, 2019, 2020 and 2021 by overlaying the respective shape-files onto the 2011 geometry and matching where at least 90% of newer ward lay inside the 2011 one
4. Using the School Information Service²⁴, educational establishment post-codes could be identified on the basis of school or education facility names, and thus combined with data from the August 2011 NSPL to derive ward code information

The final sample covering the years 2001-2019 contains 180,915 projects.

²⁴Downloaded from <https://www.get-information-schools.service.gov.uk/Downloads> on 06/04/2023

A.1.3 Sample Restrictions

To reduce noise and avoid bias, we introduce two sample restrictions on our funded project data. Firstly, we drop projects worth less than £1,000 as some projects are recorded as having received funds below the permitted lower bound of £300, thus avoiding possible measurement error problems. Moreover, many of these projects are too small to have any non-negligible effect on the crime rate. Secondly, to reduce noise and make sure that our estimates reflect youth-related projects only, we restrict our attention to those that affect youths and young people, or could benefit them in some way. To do so, we use a classification based on the text data present in the description of each project, and, as such, the approach is deterministic and rule-based. This is done to prioritise transparency and reproducibility: every classification decision can be traced directly to the presence or absence of specific patterns in the underlying text. Table A.2 below illustrates the words and phrases used in this classification.

Table A.2: Keywords for Project Classification

Strength	Keywords
Exclusion	babies, infants, toddlers, newborns, neonatal, under-5s, 0-to-5, early years, nursery, playgroup, play group, mother-and-baby, children under five/5, aged under 5, under the age of five, elderly, older people, older adults, older person, older residents, senior citizens, seniors, Age Concern, Age UK, care home, nursing home, hospice, dementia, Alzheimer's, over-60s, over-65s, over-70s, pensioners, retired, nationwide, England-wide, UK-wide, across England, across the UK, across Britain, across the country, national project / programme / initiative / scheme, world war, WWII, WW2, war veteran, armed forces veteran, ex-serviceman, Heroes Return

Continued on next page...

Strength	Keywords
Inclusion	youth, young people, young person, young adults, young men, young women, young carers, young offenders, teenagers, teens, adolescents, juveniles, pupils, schoolchildren, primary school, secondary school, high school, junior school, after-school, out-of-school/hours, extra-curricular, school-age, sixth form, Scouts, Guides, Brownies, Cubs, Cadets, Sea Cadets, Army Cadets, Girls' Brigade, Boys' Brigade, youth club, youth group, youth centre, youth project, youth worker, youth service, junior club, junior section, junior football, junior rugby, junior cricket, holiday club/programme/scheme/camp/activities, summer club/programme/scheme/camp/activities, breakfast club, mentoring, apprentice, under-10s through under-19s (e.g. U10, U16, U18) Sport England, sports hall, sports centre, sports ground, pitch, court, pavilion, complex, arena, recreation ground/centre/area, community sport, community recreation, playground, play area, skatepark, MUGA, multi-use games area, astroturf, all-weather pitch, changing room/facilities/provision, sports club, football, cricket, rugby, basketball, netball, volleyball, gymnastics, swimming, athletics, boxing, karate, judo, badminton, tennis, hockey, cycling, dance, martial arts, community centre, community center, community hall, community hub, village hall, Big Local, children, child, kids, boys, girls, students

The first tier consists of keywords, any one of which is sufficient to disqualify a record from further consideration, regardless of whatever inclusion signals may also be present. These can be loosely grouped into four categories. First, projects oriented towards infants and pre-school children, a population whose developmental needs are sufficiently distinct from those of school-age children and young people as to warrant separate treatment. Second, projects whose beneficiaries are older adults, identified through references to care settings, age-specific organisations, and conditions associated with later life such as dementia. Third, projects operating at a national or England-wide scale on the grounds that such initiatives are unlikely to represent geographically bounded, community-level provision. Finally, the projects associated with military service history, which fall outside the scope of youth-focused funding.

For records that are not excluded, we use keywords that indicate that a project directly or indirectly targets youths and young-people. First, we capture direct references, using words such as ‘youth’, ‘teenager’, ‘adolescent’, and related variants that unambiguously designate a young-person beneficiary group. Second, we include projects operating within or adjacent to formal schooling, including references to pupils, school types, and structured out-of-hours provision. Third, we include projects for established youth-sector groups such as scouts, guides, youth clubs and junior sporting sections, and those that support programme types that are strongly associated with young people even in the absence of explicit age references, including holiday and summer schemes, breakfast clubs, mentoring provision, and junior age-band competitions (e.g., U14, U16). In addition to these, we also consider keywords for project records that, while not containing explicit youth terminology, describe contexts in which young people frequently participate alongside the wider community. These mostly cover sports facilities, specific sporting disciplines, community facilities, and general references to children.

Note that if the description contains none of the keywords suggested above, it is by default placed in the “non-youth” category. As a result of this approach, 48% of projects are classed as “youth” projects.

A.1.4 Project Classifications

In addition to the above, we classify the youth projects thematically. This is done in two ways. The first focuses on their goals and objectives: Health & Well-being, Skills & Opportunities, Civil & Social Engagement and Other. Table A.3 provides the list of keywords used for this classification. “Health & Well-being” projects are those that provide opportunities for physical activity or outdoor activities, organise mental health support & therapy or aim to improve the general well-being of young people. “Skills & Opportunities” projects are those that provide additional education opportunities and training, support learning and enterprise, or seek to develop creative and digital skills. Next, “Civil & Social Engagement” projects encourage community action and civic participation, and seek to raise awareness around the issues concerning the environment and sustainability. The final type, “Other” is for all other projects that cannot be placed into any of the other three groups.

Table A.3: Project Classification Keywords

Theme	Keywords
Health & Well-being	sport(s), fitness, exercise, physical activity, physical health, swimming, cycling, walking, running, yoga, mental health, counselling, counsellor, therapy, therapeutic, therapist, mindfulness, wellbeing, well-being, well being, health, nutrition, obesity, healthy eating, play, recreation, outdoor (activities / education), adventure (programmes / play), palliative, respite (care)
Skills & Opportunities	education, educational, tutoring, tutors, STEM, science, technology, engineering, maths, literacy, numeracy, employability, employment, jobs, career, apprentice, entrepreneur, enterprise, digital skills, coding, technology, art(s), music, creative, creativity, craft, dance, drama, theatre, film, media, training
Civil & Social Engagement	volunteering, volunteers, advocacy, youth council, community building, social action, environmental, conservation, sustainable, sustainability, civic, citizenship, democracy, democratic
Other	Records not matching any pattern in categories 1–3 above

Summary statistics for this first thematic classification are provided in Table A.4. Most projects, both large and small, belong to the “Health & Well-being” category.

Table A.4: Proportion of Projects by Type and Size

Project Type	Small Projects	Large Projects
Civil & Social Engagement	0.019	0.024
Health & Well-being	0.798	0.515
Other Activities	0.100	0.260
Skills & Opportunities	0.083	0.200

^a Note: Values are given to the nearest pound sterling. Projects over £10,000 are classified as Large; those worth £10,000 or less are Small.

The second thematic classification focuses on the type of each project: “Activities & Services”, “Assets & Infrastructure ” and “Other”. The keywords used for this classification are given in Table A.5. The crucial distinction is between the first two groups. The former contains projects that fund events and gatherings usually with one-off or short-term contact. The latter concerns projects that support equipment purchases, building refurbishments and renovations that may have a more long-lasting impact. As before, the final group, “Other” is for all other projects that cannot be placed into any of the other groups.

Table A.5: Project Classification Keywords

Category	Keywords
Assets & Infrastructure	refurbish(ment), renovation, renovating, build + centre / hall / facility / pitch / court / pool / playground / arena / studio / hub; construct + centre / hall / facility / pitch / court / pool; new pitch / court / pool / hall / centre / studio / arena / ground / playground, install + equipment / trail / pitch / court / flooring / lighting / fencing / shed / shelter, purchase + equipment / kit / gear / instrument / vehicle / minibus / bus / transport, upgrade + facility / pitch / court / lighting / studio, capital grant / investment / project / fund, durable equipment, IT suite, sports gear, new equipment, new flooring, new furniture
Activities & Services	workshop(s), programme(s), program, session(s), club(s), class(es), course(s), activity / activities, event(s), scheme, deliver (sessions / a programme), service(s), outreach, provide support, provide care, coaching, mentoring, counselling
Other	Records not matching any pattern in categories 1–2 above

Table A.6 presents summary statistics for this second classification. Interestingly, “Activities & Services” are the most common type, independent of the size of the project.

Table A.6: Proportion of Projects by Type and Size

Project Type	Small Projects	Large Projects
Activities & Services	0.602	0.664
Assets & Infrastructure	0.305	0.143
Other	0.092	0.193

^a Note: Values are given to the nearest pound sterling. Projects over £10,000 are classified as Large; those worth £10,000 or less are Small.

A.2 Local Crime Data

A.2.1 Sources

The main source of crime data for the analysis is the police reported data available at the Police Data Archive. Further details available in Table A.7 below.

Table A.7: Raw Crime Data Sources

Name	Link
Police Recorded	Source

Notes: This table reports the type of crime data used and the link to where it can be accessed.

A.2.2 LSOA to Ward Aggregation

The LSOA-level police recorded data is aggregated to the ward-level. However, there is no publicly available "best-fit" look-up table for LSOAs to Wards (2011). Therefore, one has been constructed by matching the population weighted centroids of LSOAs to the fully clipped 2011 census ward boundaries. Geographically, LSOAs are not constructed in such a way as to aggregate into wards. Hence, the matching process is imperfect. This is because, though there are 8,546 census merged wards, LSOA population weighted centroids fall into 8,535 of them. This means that there are 11 wards such that no LSOA population centroids are inside them. This has potential to introduce some spatial-based

measurement error into the aggregated crime data in that some counts of crime are misattributed geographically. Prima facie, this may appear to bias results since it implies that some wards where projects have been undertaken will not be observed. However, the population weighted centroid adjusts for the distribution of the population across each LSOA, and so areas which are ‘discarded’ in the matching process are unlikely to have a significant number of occupants.

A.2.3 Crime Classifications

The raw police recorded crime data is summarised below. It is worth noting the changes that took place in some of these categories that explain the differences in the number of observations between each one. Firstly, prior to September 2011, the ‘arson and criminal damage’ was part of the ‘other crime’ category, and was not recorded independently. Secondly, in June 2013, the ‘public order and weapons’ category was split into ‘public order’ and ‘possession of weapons’ categories. Thirdly, in the same month, the ‘bike theft’ and ‘theft from person’ categories were introduced out of the ‘other theft’ category.

As stated in the main text these raw values are aggregated into six broader groups. Hence, ‘total crime’ is simply the sum across all crime types except ‘other crime’; ‘total crime without ASB’ is the sum of all crime excluding anti-social behaviour and other crime; ‘property crime’ includes burglary, shoplifting, vehicle crime, bike theft, theft from person and other theft; ‘violent crime’ includes violent & sexual offences, robbery, arson, weapons, public order and disorder incidents, while ‘substance crime’ is kept unchanged. Note that the values reported in Table A.8 below are for the full sample before any restrictions are based due to changing counting rules by the Home Office.

Table A.8: Raw Crime Categories Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Anti-social Behaviour	920,490	2.45	2.64	0.00	100.00
Violent & Sexual	920,490	1.37	1.57	0.00	149.31
Other Theft	920,490	0.63	1.62	0.00	237.47
Arson & Criminal Damage	877,815	0.67	0.69	0.00	23.11
Burglary	920,490	0.54	0.56	0.00	32.38
Public Order	672,975	0.33	0.57	0.00	40.76
Vehicle	920,490	0.46	0.53	0.00	27.23
Drugs	920,490	0.18	0.50	0.00	121.33
Bike Theft	672,975	0.10	0.33	0.00	34.56
Robbery	920,490	0.06	0.17	0.00	14.21
Theft from Person	672,975	0.08	0.68	0.00	72.28
Public Disorder and Weapons	247,515	0.12	0.33	0.00	25.52
Possession of Weapons	672,975	0.04	0.13	0.00	14.43
Other Crime	920,490	0.31	1.39	0.00	282.55

A.3 Additional Variables

Control variables were selected to represent various dimensions of local-area development such as wealth, education, extent of urbanisation and poverty.

Table A.9 provides a brief explanation of each variable.

Table A.9: Control Variable Descriptions

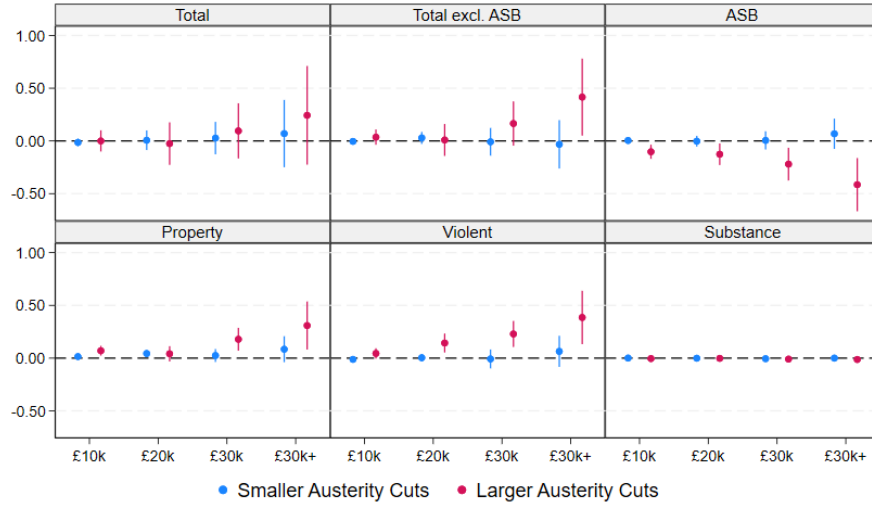
Variable	Explanation
Population	The number of residents of a ward.
Youth Share	The share of residents aged 0–24.
L2 Education Share	The share of individuals who have achieved at least an L2-level (GCSE) of education.
Social Rent Share	Share of individuals living in socially-rented accommodation.
Urban/Rural	ONS classification of whether a given ward is urban or rural.
Austerity Loss	Estimated per person loss (GBP) from austerity cuts calculated by Beatty and Fothergill (2013). Note that for some local authorities, these values are not available.

B Online Appendix

B.1 Additional Results

In this section, we briefly discuss results that did not make into the main text, but are important for how we interpret the effectiveness of each type of project. Specifically, we investigate the interaction between the size of austerity cuts and the quantity of funding active in a ward. The 2012 UK Welfare Reform Act introduced deep cuts to the local provision of public services, including community centres and local clubs, which caused increases in local crime (Giulietti and McConnell, 2020; Villa, 2024). In areas more affected by austerity measures, funding injections may partially replace lost public youth services, leading to a larger response in these areas. We estimate the effects of additional funding in areas where the cost of austerity was below respectively above the median following the start of the reform in April 2013, shown in Figure B.1 below.²⁵

Figure B.1: Marginal Effects of Micro Projects by Austerity Costs



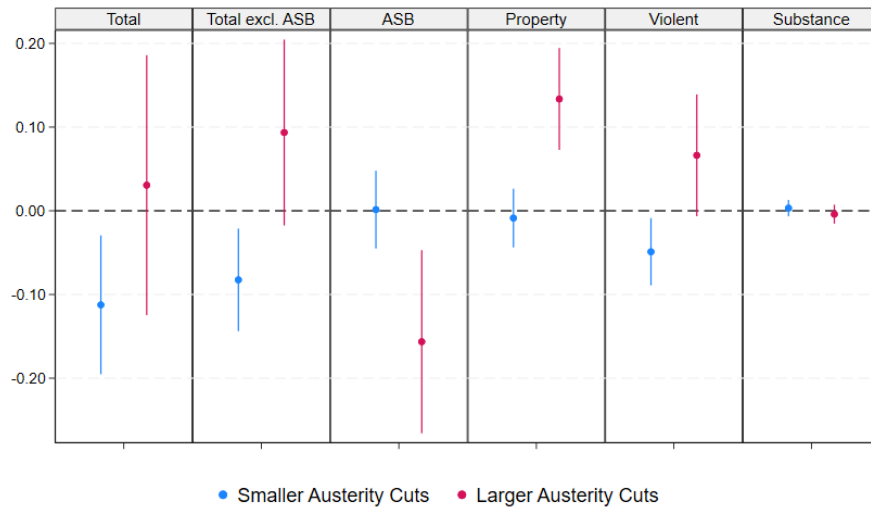
Notes: A plot of treatment effects estimates for the impact of receiving different quantities of funds for micro projects on crime by whether a ward is affected by austerity policies (cuts are below or above the median). A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

²⁵The estimates along the extensive margin are given in Figure B.6 in Appendix B.

We find that projects have greater effects in areas that saw larger austerity cuts. Micro projects in areas strongly impacted by austerity cuts reduce anti-social behaviour. Yet, they also increase both property and violent crime. The changes are commensurate across all three types of crime at around 15%.

We now consider the relationship between large projects and austerity. As before, due to cuts in public expenditures, many opportunities for young people had been eliminated. Large projects are those that, alongside the numerous smaller interventions, may act to mitigate these costs. This is illustrated in Figure B.1

Figure B.2: Marginal Effects of Large Projects by Austerity Costs



Notes: A plot of treatment effects estimates for the impact of receiving at least one large project on crime by whether a ward is affected by austerity policies (cuts are below or above the median). A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

Here we find an interesting contrast: with the exception of anti-social behaviour (ASB), areas unaffected by austerity see reductions in crime, while those more strongly impacted by cuts saw growth! It is important to note that this is partly consistent with the micro project estimates where property and violent crime rise in high austerity neighbourhoods. The key lesson for policymakers appears

to be that these social projects are best thought of as complements to existing activities, centres and opportunities, and not their substitutes. Although both small and large projects have some incapacitation effects in high austerity neighbourhoods, it is by improving existing organisations that a difference can be made.

B.2 Additional Tables

Table B.1: Number of Active Projects Estimates

	(1) Total	(2) Total Excl. ASB	(3) ASB	(4) Property	(5) Violent	(6) Substance
1	-0.007 (0.015)	0.006 (0.012)	-0.014 (0.009)	0.009 (0.006)	-0.004 (0.008)	0.001 (0.002)
2	0.019 (0.033)	0.036 (0.023)	-0.030* (0.017)	0.033*** (0.011)	0.024 (0.015)	0.001 (0.003)
3	0.021 (0.048)	0.045 (0.038)	-0.077*** (0.029)	0.041** (0.019)	0.027 (0.027)	-0.003 (0.004)
3+	0.028 (0.093)	0.082 (0.078)	-0.153*** (0.048)	0.074** (0.037)	0.109** (0.055)	-0.004 (0.006)
Mean	7.167	4.635	2.474	2.160	2.199	0.179
Rural	4.124	2.690	1.402	1.312	1.231	0.093
Urban	8.639	5.577	2.994	2.571	2.668	0.221
Percentile	50.293	51.238	46.009	52.263	52.979	0.000
N	353765	353765	491646	491646	353765	491646

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the two-stage difference-in-difference estimation results illustrating the average treatment effect of receiving different numbers of community micro-projects on the monthly crime rate at the ward-level. The crime rates running from models (1) to (6) are total crime, total crime excluding anti-social behaviour (ASB), anti-social behaviour (ASB), property crime, violent crime and substance crime respectively. The mean values for each type of crime, and their values in rural and urban wards are also reported alongside the change in the percentile of the crime distribution that a ward would see given the estimated treatment effect. Standard errors are given in parentheses and are clustered at the ward-level.

Table B.2: Share of Project Types by Size

Type	Micro	Large
Classification 1		
Civil & Social Engagement	2.1%	2.4%
Health & Well-being	70.3%	50.2%
Other Activities	16.7%	26.1%
Classification 2		
Skills & Opportunities	11.0%	21.3%
Activities & Services	59.7%	68.3%
Assets & Infrastructure	25.5%	13.3%
Other	14.9%	18.4%

Table B.3: Micro-project and Local Crime by Project Type I

	(1) Tot.	(2) Tot. Excl. ASB	(3) ASB	(4) Prop.	(5) Viol.	(6) Subst.
Panel A: Extensive Margin						
H& W	0.003 (0.019)	0.020 (0.014)	-0.020* (0.011)	0.022*** (0.008)	0.007 (0.009)	-0.001 (0.002)
S& O	-0.029 (0.040)	-0.019 (0.029)	-0.032 (0.024)	0.035** (0.016)	-0.013 (0.022)	-0.000 (0.003)
C& S	0.205*** (0.080)	0.165*** (0.063)	0.063 (0.044)	-0.028 (0.030)	0.110*** (0.034)	0.014 (0.019)
Other	0.008 (0.025)	0.018 (0.019)	-0.041*** (0.014)	0.005 (0.009)	0.016 (0.013)	0.002 (0.002)
Panel B: Intensive Margin						
H & W	0.011 (0.017)	0.028** (0.014)	-0.022** (0.009)	0.018** (0.007)	0.019** (0.009)	-0.002 (0.001)
S & O	-0.042 (0.040)	-0.027 (0.032)	-0.046* (0.025)	0.027 (0.017)	-0.020 (0.024)	-0.000 (0.004)
C & S	0.235*** (0.090)	0.195*** (0.069)	0.071 (0.053)	-0.030 (0.034)	0.122*** (0.039)	0.016 (0.019)
Other	0.021 (0.028)	0.028 (0.022)	-0.046*** (0.015)	0.016 (0.010)	0.025* (0.015)	0.003 (0.002)
Mean	7.167	4.635	2.474	2.160	2.199	0.179
Rural	4.124	2.690	1.402	1.312	1.231	0.093
Urban	8.639	5.577	2.994	2.571	2.668	0.221
N	353765	353765	491646	491646	353765	491646

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the two-stage difference-in-difference estimation results illustrating the average treatment effect of receiving different types of micro-projects on the monthly crime rate at the ward-level. Panel A examines projects along their extensive margin, capturing the effect of having at least one project of a particular type active. Panel B contains estimates along the intensive margin based on the total sum of funding active for projects of each type. There are four types of projects considered here depending on their focus: (i) Health and Wellbeing, (ii) Skills and Opportunities, (iii) Civic and Social Engagement, and (iv) Other. The crime rates running from models (1) to (6) are total crime, total crime excluding anti-social behaviour (ASB), anti-social behaviour (ASB), property crime, violent crime and substance crime respectively. The mean values for each type of crime, and their values in rural and urban wards are also reported. Standard errors are given in parentheses and are clustered at the ward-level.

Table B.4: Micro-project and Local Crime by Project Type II

	(1) Tot.	(2) Tot. Excl. ASB	(3) ASB	(4) Prop.	(5) Viol.	(6) Subst.
Panel A: Extensive Margin						
A & I	0.004 (0.028)	0.016 (0.022)	-0.003 (0.015)	0.002 (0.010)	0.024 (0.015)	-0.002 (0.003)
A & S	0.004 (0.021)	0.017 (0.016)	-0.027** (0.012)	0.021*** (0.008)	0.006 (0.011)	0.003 (0.002)
Other	0.002 (0.025)	0.013 (0.019)	-0.030** (0.015)	0.021** (0.011)	-0.000 (0.013)	-0.004 (0.002)
Panel B: Intensive Margin						
A & I	0.018 (0.033)	0.031 (0.027)	-0.003 (0.016)	0.003 (0.011)	0.038** (0.018)	-0.002 (0.003)
A & S	0.012 (0.020)	0.028* (0.016)	-0.038*** (0.011)	0.022*** (0.007)	0.019* (0.011)	0.001 (0.002)
Other	0.005 (0.027)	0.010 (0.020)	-0.022 (0.016)	0.015 (0.011)	-0.001 (0.013)	-0.003 (0.002)
Mean	7.167	4.635	2.474	2.160	2.199	0.179
Rural	4.124	2.690	1.402	1.312	1.231	0.093
Urban	8.639	5.577	2.994	2.571	2.668	0.221
N	353765	353765	491646	491646	353765	491646

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the two-stage difference-in-difference estimation results illustrating the average treatment effect of receiving different types of micro-projects on the monthly crime rate at the ward-level. Panel A examines projects along their extensive margin, capturing the effect of having at least one project of a particular type active. Panel B contains estimates along the intensive margin based on the total sum of funding active for projects of each type. There are three types of projects considered here depending on their focus: (i) Assets and Infrastructure, (ii) Activities and Services, and (iii) Other. The crime rates running from models (1) to (6) are total crime, total crime excluding anti-social behaviour (ASB), anti-social behaviour (ASB), property crime, violent crime and substance crime respectively. The mean values for each type of crime, and their values in rural and urban wards are also reported. Standard errors are given in parentheses and are clustered at the ward-level.

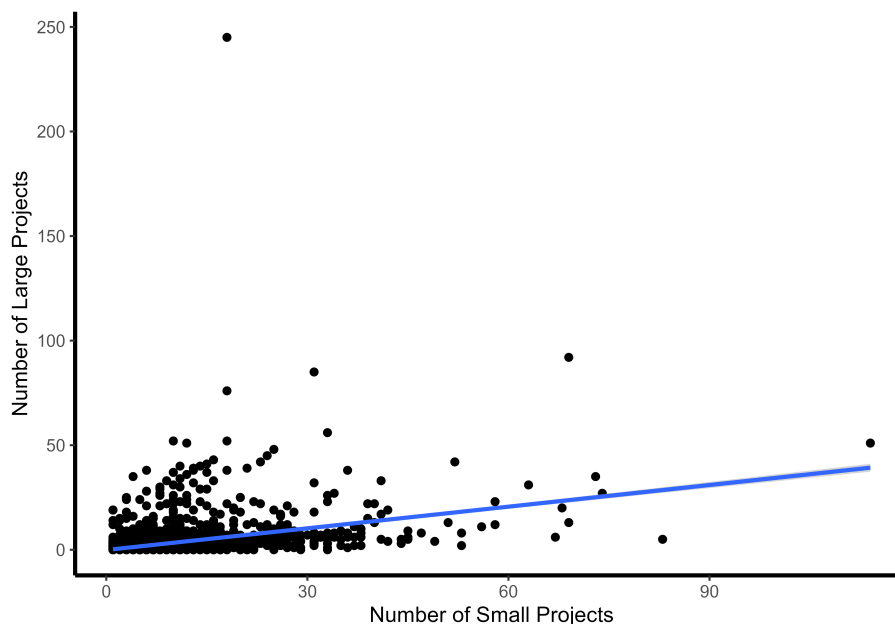
Table B.5: Extensive Margin across Large Project Types

	(1) Total	(2) Total Excl. ASB	(3) ASB	(4) Property	(5) Violent	(6) Substance
Panel A: Type 1						
H & W	-0.044 (0.044)	-0.037 (0.033)	-0.014 (0.026)	0.032* (0.016)	-0.042* (0.022)	-0.001 (0.004)
S & O	-0.299* (0.153)	-0.195 (0.121)	0.114 (0.092)	-0.050 (0.059)	-0.155* (0.094)	0.012 (0.013)
C & S	-0.343* (0.196)	-0.142 (0.129)	-0.136 (0.118)	0.012 (0.069)	-0.077 (0.073)	0.021 (0.024)
Other	-0.195 (0.176)	-0.170 (0.127)	0.084 (0.089)	0.030 (0.045)	-0.115 (0.098)	0.001 (0.017)
Panel B: Type 2						
A & I	-0.013 (0.043)	-0.017 (0.034)	0.002 (0.024)	0.018 (0.017)	-0.025 (0.022)	-0.003 (0.004)
A & S	-0.194** (0.091)	-0.074 (0.063)	-0.051 (0.055)	0.071** (0.029)	-0.064 (0.046)	0.005 (0.008)
Other	-0.054 (0.144)	-0.168 (0.123)	0.114 (0.087)	-0.113* (0.060)	-0.130 (0.109)	-0.007 (0.025)
Mean	7.167	4.635	2.474	2.160	2.199	0.179
Rural	4.124	2.690	1.402	1.312	1.231	0.093
Urban	8.639	5.577	2.994	2.571	2.668	0.221
N	266336	266336	380335	380335	266336	380335

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the two-stage difference-in-difference estimation results illustrating the average treatment effect of receiving at least one large project of a particular type on the monthly crime rate at the ward-level. In Panel A, we consider the following project types: (i) Health and Wellbeing, (ii) Skills and Opportunities, (iii) Civic and Social Engagement, and (iv) Other. In Panel B, there are three additional types of projects: (i) Assets and Infrastructure, (ii) Activities and Services, and (iii) Other. The crime rates running from models (1) to (6) are total crime, total crime excluding anti-social behaviour (ASB), anti-social behaviour (ASB), property crime, violent crime and substance crime respectively. The mean values for each type of crime, and their values in rural and urban wards are also reported. Standard errors are given in parentheses and are clustered at the ward-level.

Figure B.3: Scatter Plot for the Number of Small and Large Projects



Notes: A scatter plot showing the correlation between the number of small and large projects wards receive in the 2011-2019 period.

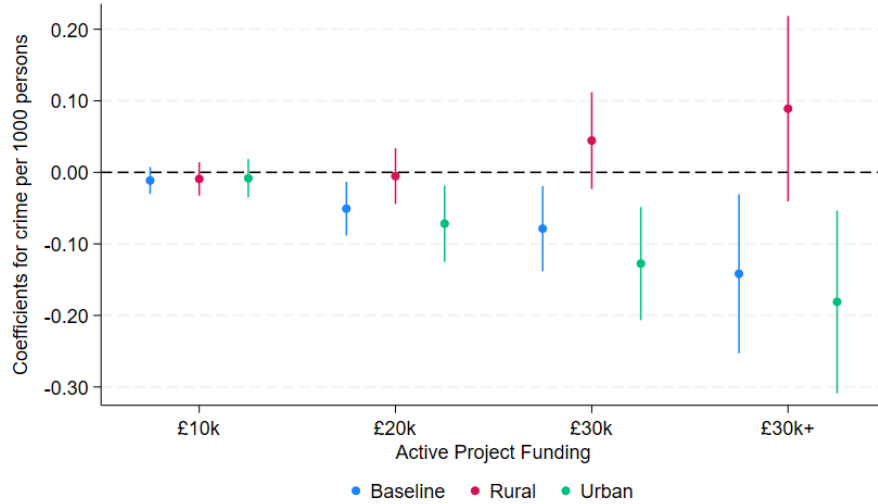
Table B.6: Extensive Margin for Large Projects - Spatial Spillovers

	(1) Tot.	(2) Tot. Excl. ASB	(3) ASB	(4) Prop.	(5) Viol.	(6) Subst.
Large	-0.068* (0.038)	-0.076*** (0.028)	0.013 (0.022)	0.032** (0.014)	-0.086*** (0.020)	0.002 (0.003)
Neigh. Micro	-0.026 (0.021)	-0.010 (0.015)	-0.012 (0.015)	-0.002 (0.009)	0.003 (0.011)	-0.002 (0.002)
Neigh. Large	0.032 (0.023)	0.013 (0.017)	0.019 (0.016)	-0.001 (0.010)	-0.002 (0.012)	0.002 (0.002)
Mean	7.167	4.635	2.474	2.160	2.199	0.179
Rural	4.124	2.690	1.402	1.312	1.231	0.093
Urban	8.639	5.577	2.994	2.571	2.668	0.221
N	266336	266336	380335	380335	266336	380335

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the two-stage difference-in-difference estimation results illustrating the average treatment effect of receiving a large community project on the monthly crime rate at the ward-level. In addition, we include two dummies for whether a neighbouring ward is receiving a micro or large project to test for spatial spillovers. The crime rates running from models (1) to (6) are total crime, total crime excluding anti-social behaviour (ASB), anti-social behaviour (ASB), property crime, violent crime and substance crime respectively. The mean values for each type of crime, and their values in rural and urban wards are also reported. Standard errors are given in parentheses and are clustered at the ward-level.

Figure B.4: Concurrent Micro Project Estimates



Notes: A plot of two-stage difference-in-difference estimates for receiving different quantities of funding for micro projects on the rate of anti-social behaviour. The results are also split by whether wards are classed as urban or rural.

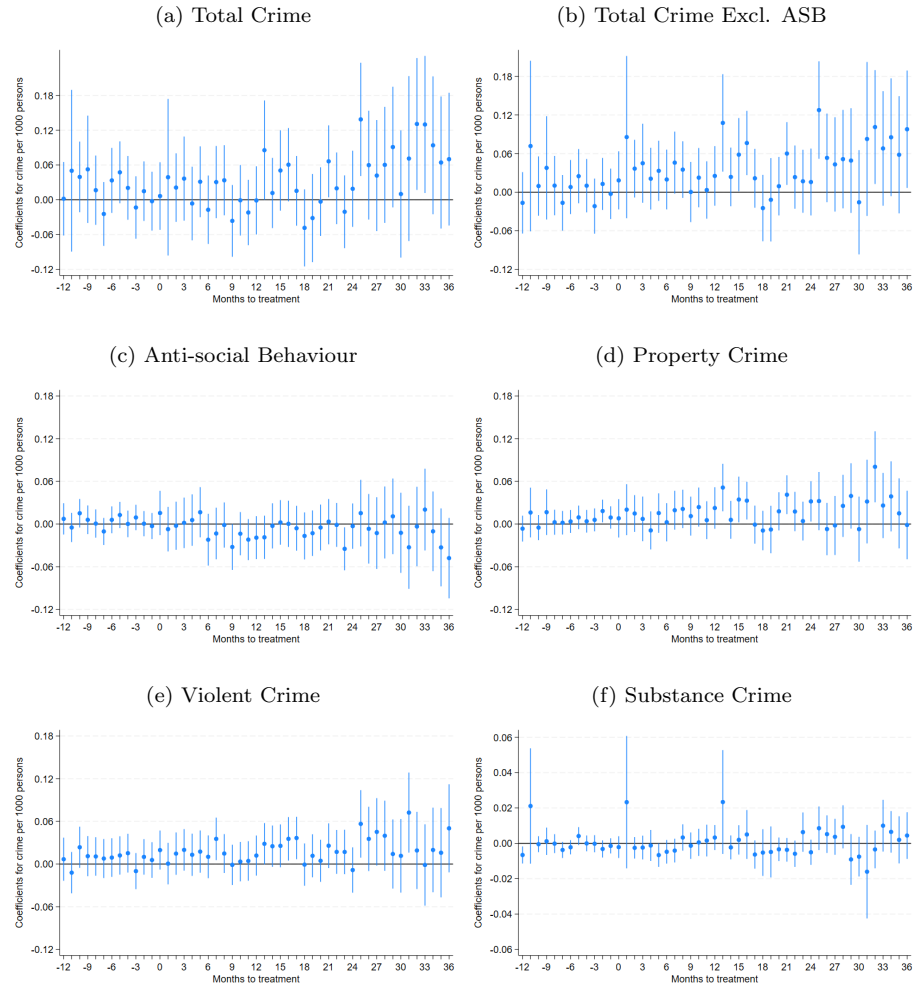
B.3 Additional Figures

Figure B.6: Effects of Micro Projects by Austerity Costs



Notes: A plot of treatment effects estimates for the impact of receiving at least one micro project on different types of crime by whether a ward is affected by austerity policies (cuts are below or above the median). A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.

Figure B.5: Event Study Plots by Type of Crime



Notes: A set of event study plots for the effect of receiving at least one micro project on different types of crime. The dependent variable is reported by type of crime per 1,000 residents. A 95% confidence interval based on standard errors clustered at the ward-level is included around the estimates.